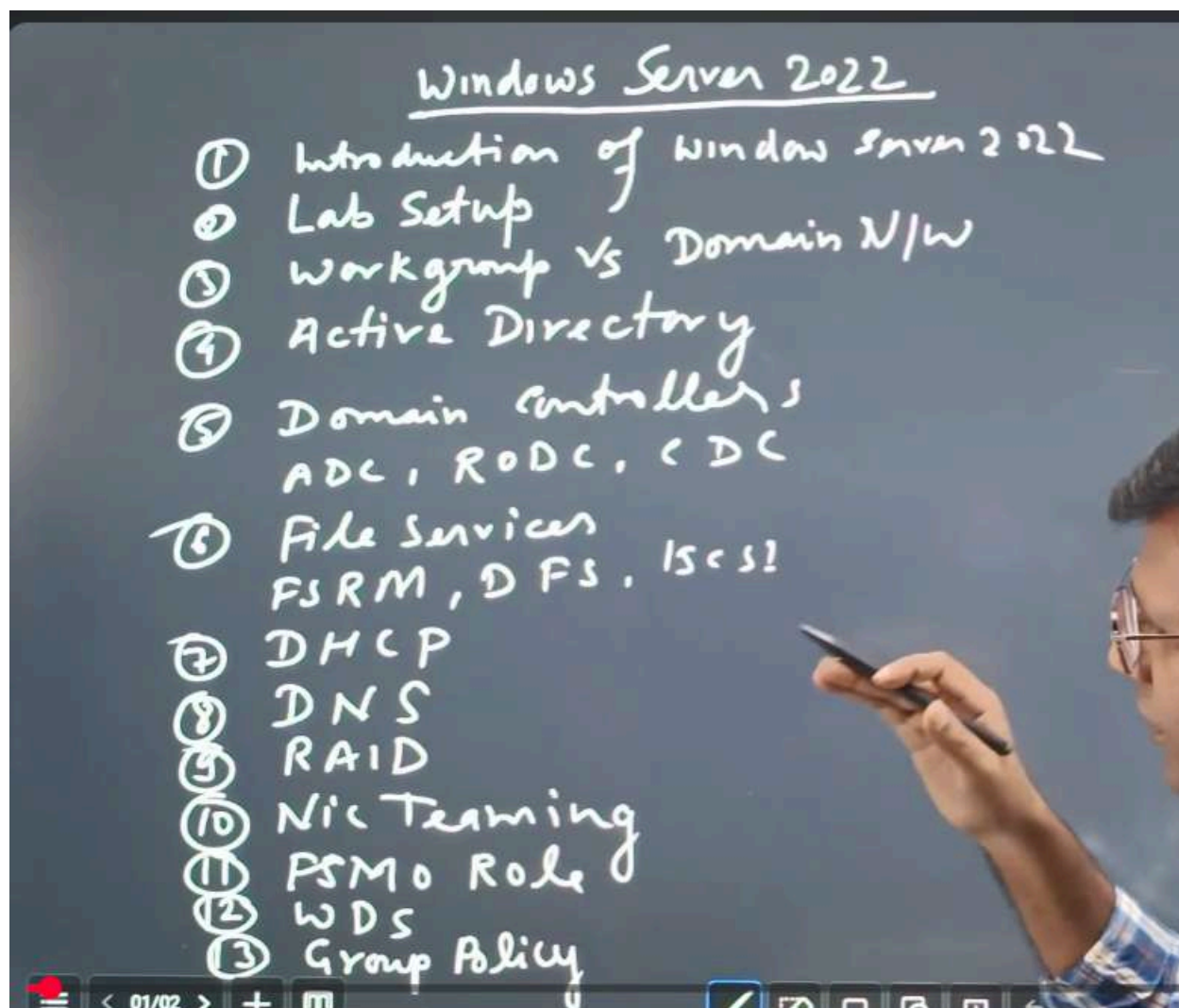
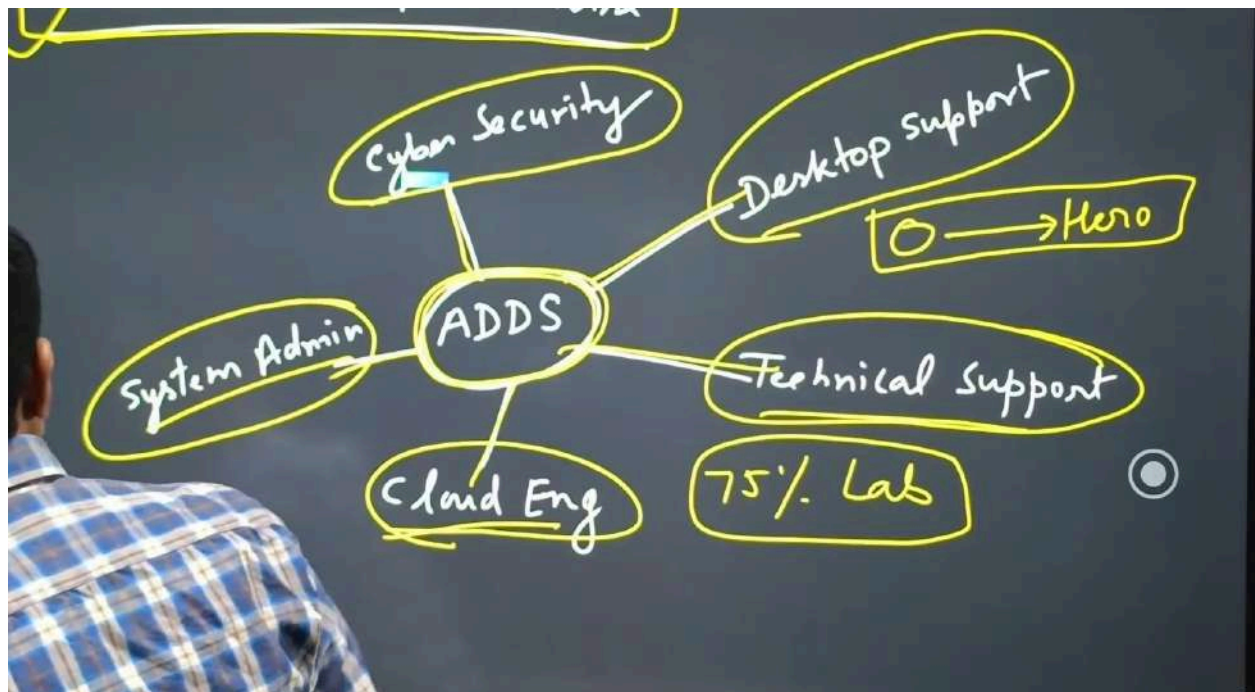


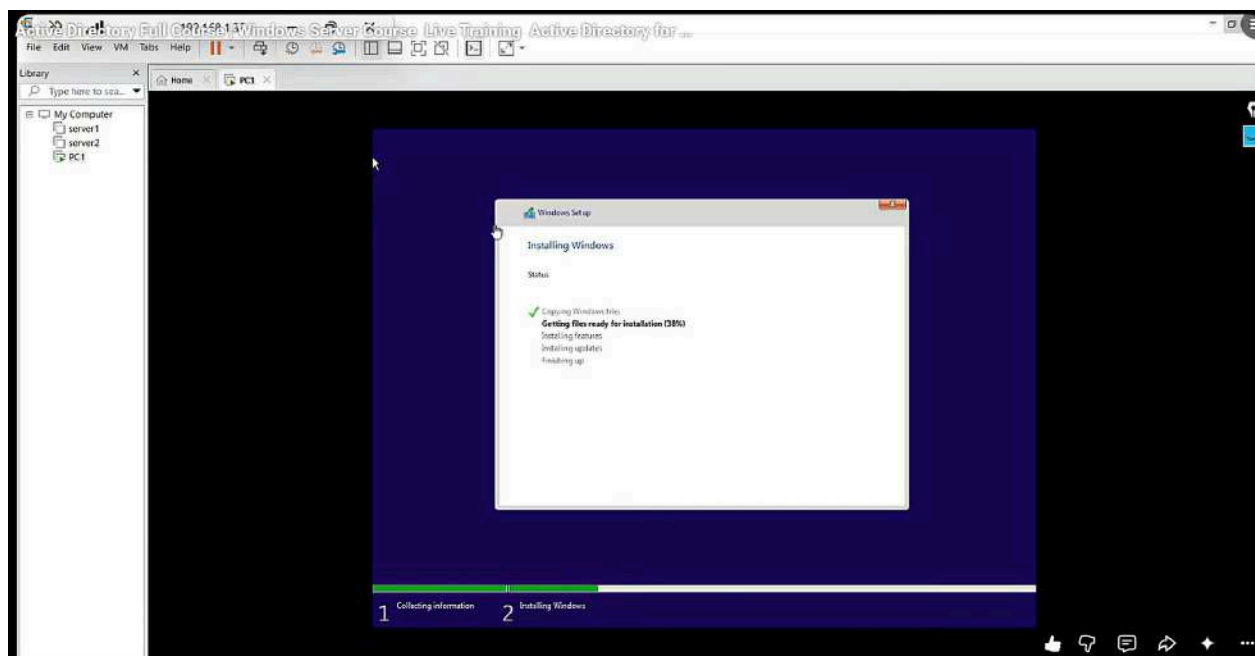
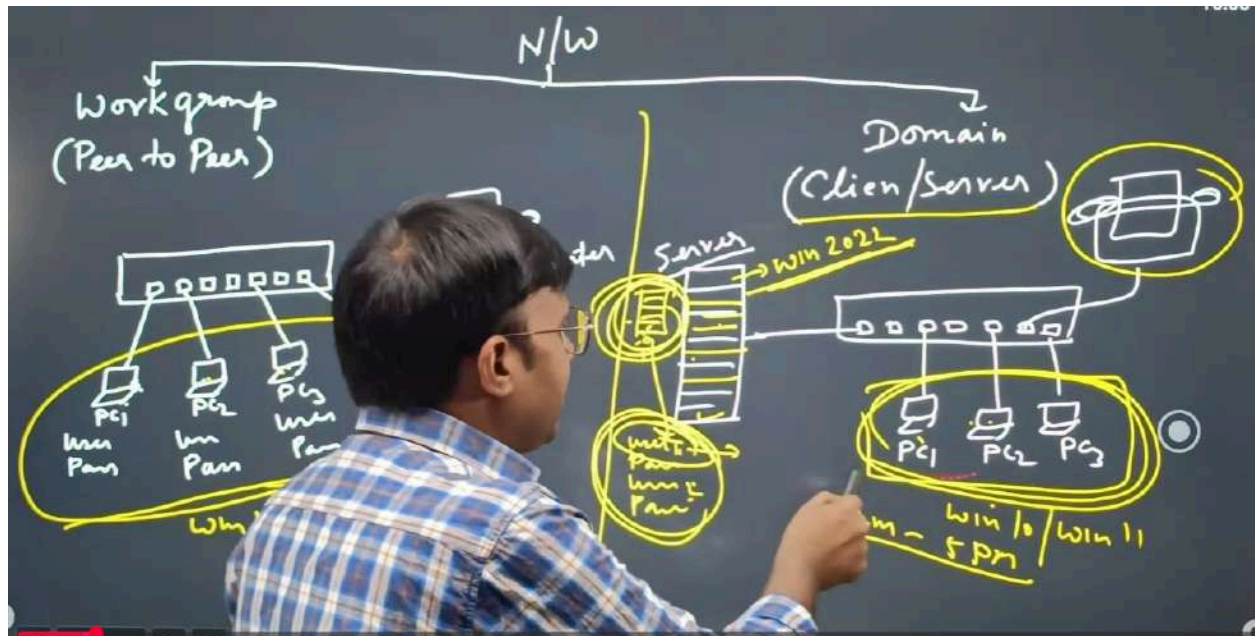


After completing this Windows Server course, I will have achieved:

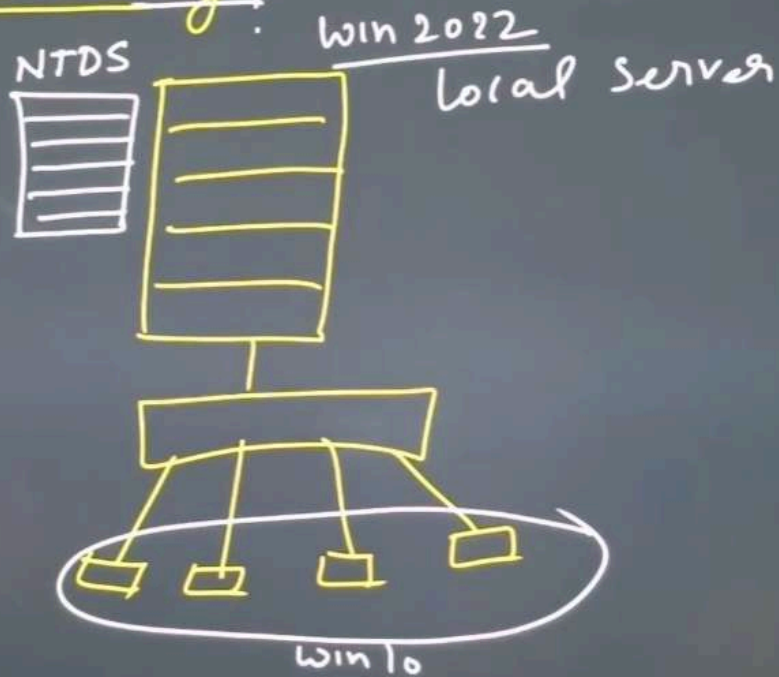
- Windows Server fundamentals
- Windows Server editions and versions
- Client vs. Server operating systems
- VMware Workstation setup
- Windows Server lab environment setup
- Windows Server 2022 installation
- Windows 10/11 client installation
- Virtual machine creation and management
- Workgroup vs. Domain networking
- Active Directory Domain Services (AD DS)
- Active Directory installation
- Domain joining
- User account management

- Group management
- Organizational Unit (OU) management
- Additional Domain Controller (ADC)
- Read-Only Domain Controller (RODC)
- Child Domain Controller
- File Server Resource Manager (FSRM)
- Distributed File System (DFS)
- iSCSI Initiator
- DHCP configuration
- DNS configuration
- Disk management
- Basic and Dynamic disks
- RAID configuration (RAID 1, RAID 5, RAID 6, RAID 10)
- NIC Teaming
- FSMO roles
- Active Directory replication
- Windows Deployment Services (WDS)
- Group Policy (GPO) management
- User and computer policy configuration
- Windows Server dashboard and Server Manager
- VMware Tools installation
- Windows Server hands-on labs
- End-to-end Windows Server administration workflow





Active Directory?



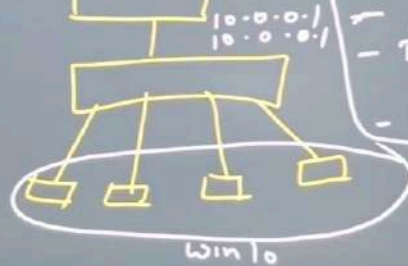
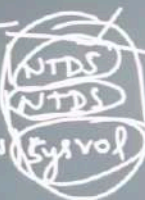
Active Directory?

Win 2022

Local Server

Location

C:\Windows
C:\Windows
C:\Windows



Prerequisite

- Server OS
- Static IP
- NTFS Partition
- Registered Domain
intacademy.in
- DNS

11:53

Active Directory?

Location
NTDS
win 2022
11:58

c:\window NTDS

c:\window NTDS

c:\window NTDS

Local Server

10.0.0.1

10.0.0.1

10ntacademy\administrator

administrator@10ntacademy.in

SAM

compmgmt.msc

Active Directory

win Server 2022
12:14

Sales Dept

100 user

IT Dept

100 user

User 1

Pass

User 2

Pass

User 3

Pass

10ntacademy.m

Ad objects

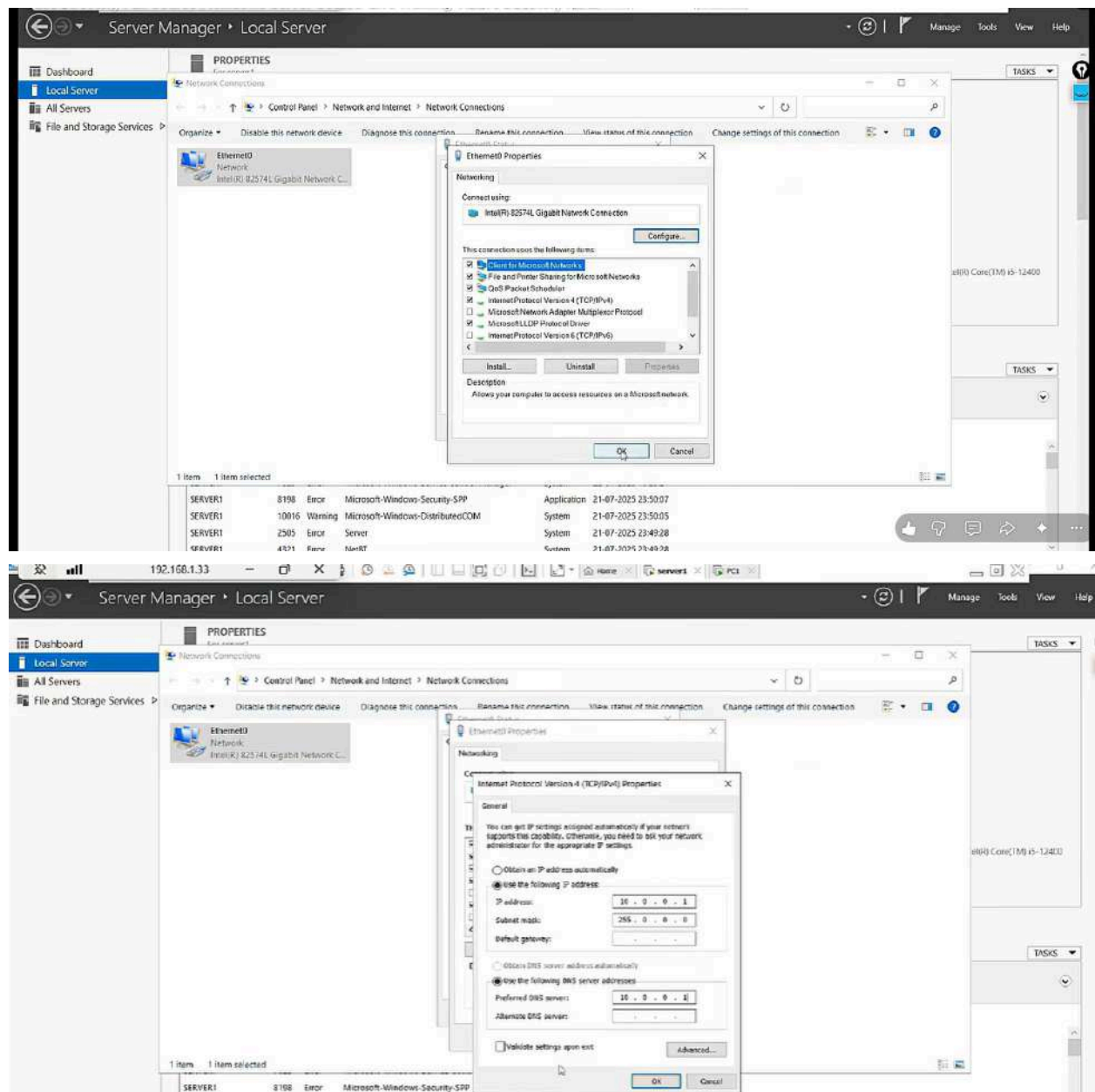
Users

Groups

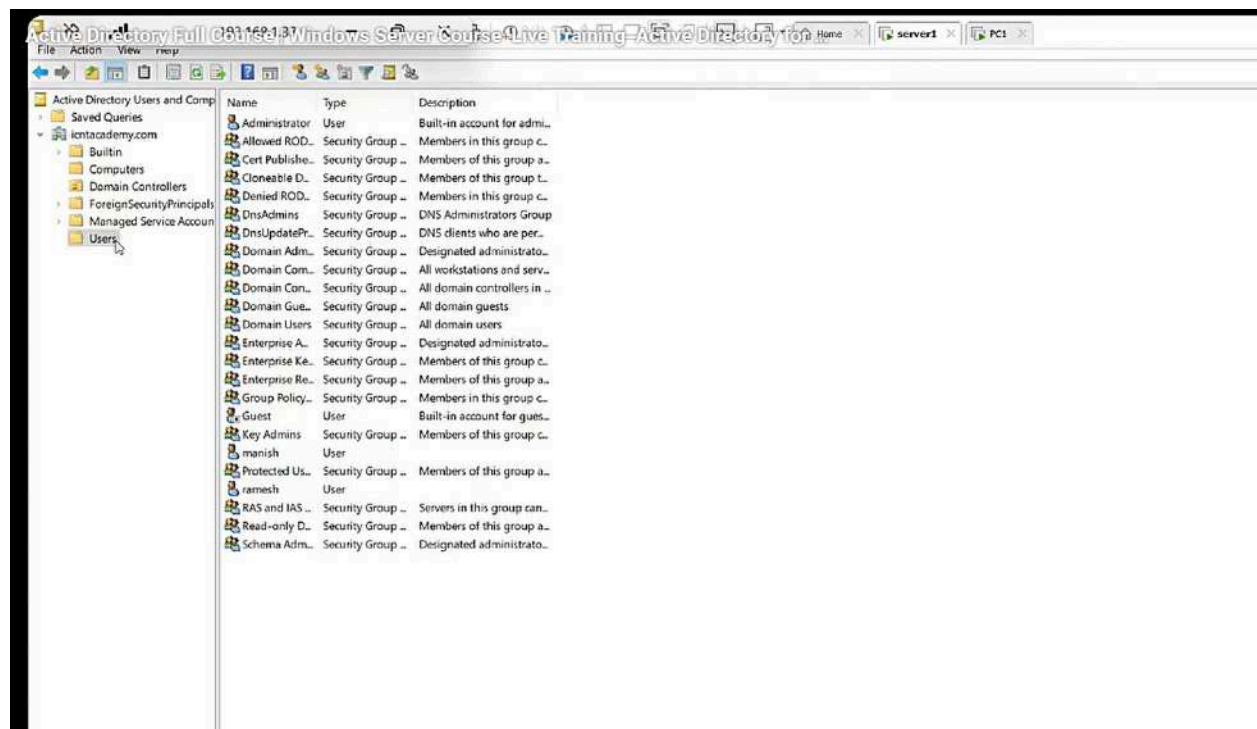
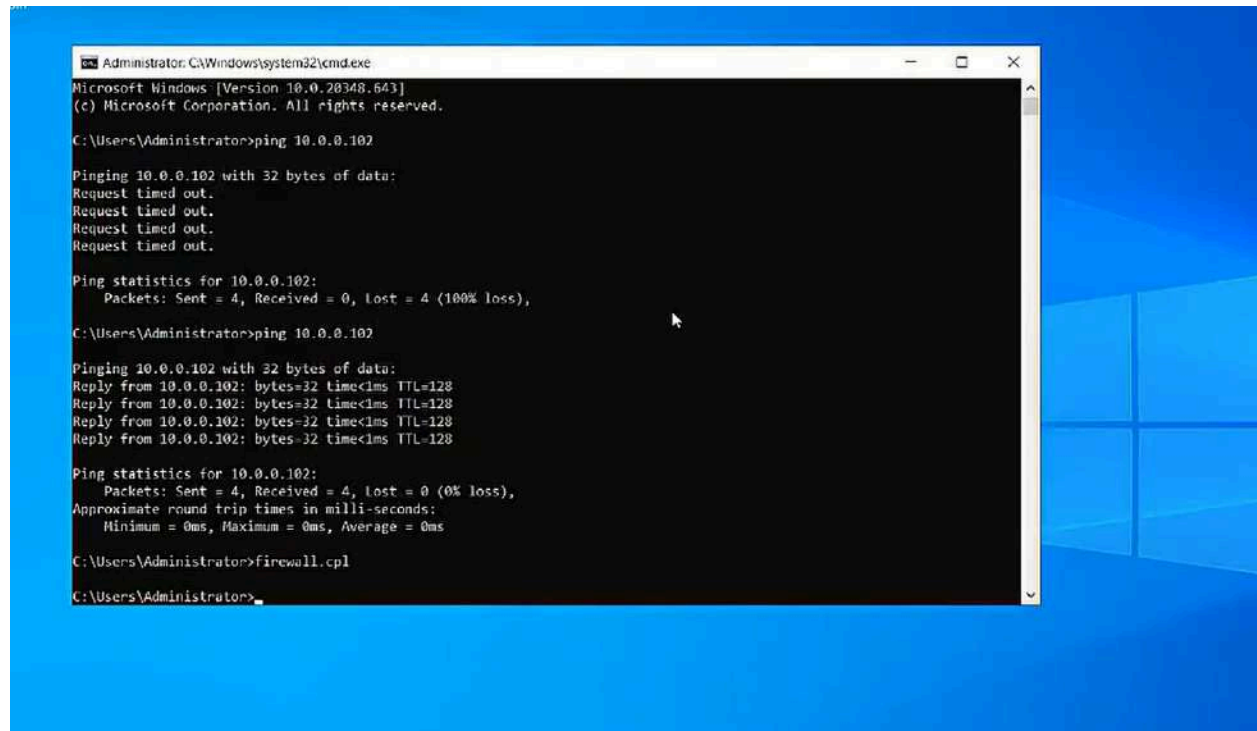
Computer

OU

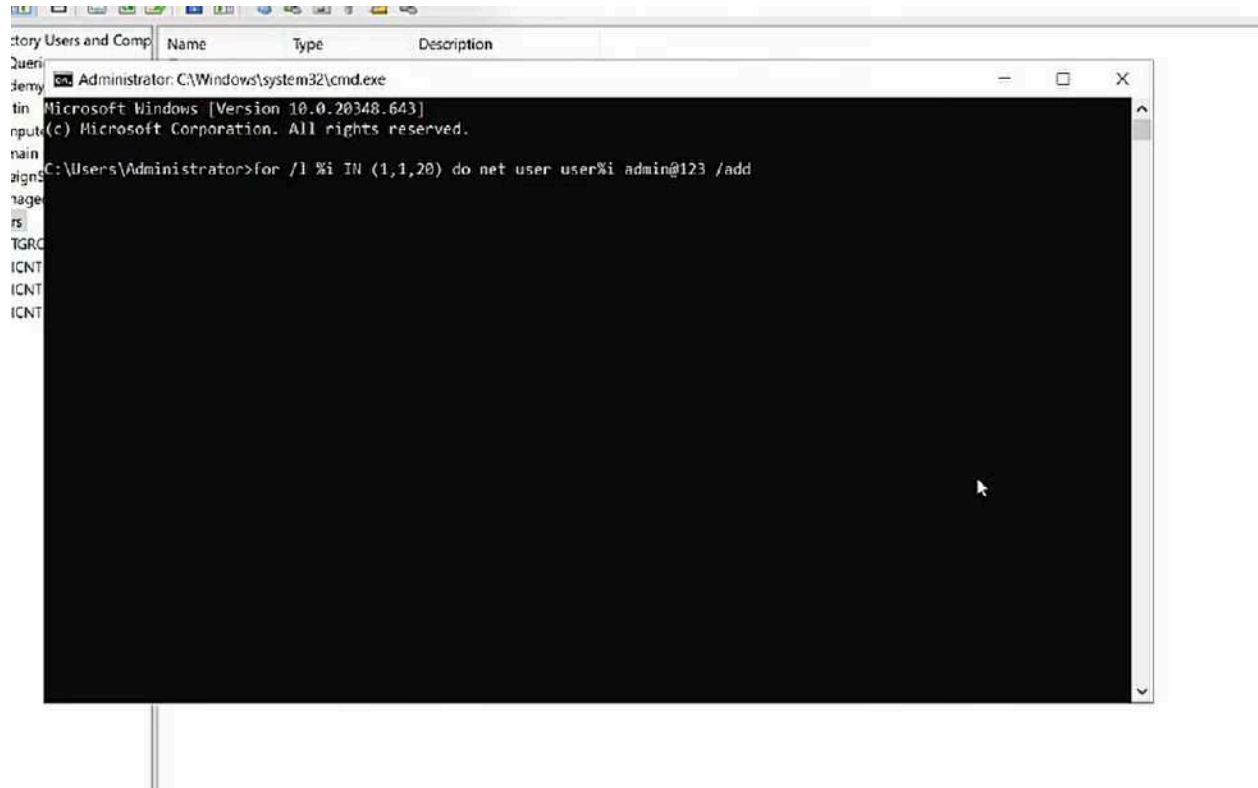
PC, P2



Firewall off for both server and Client Pc connection:



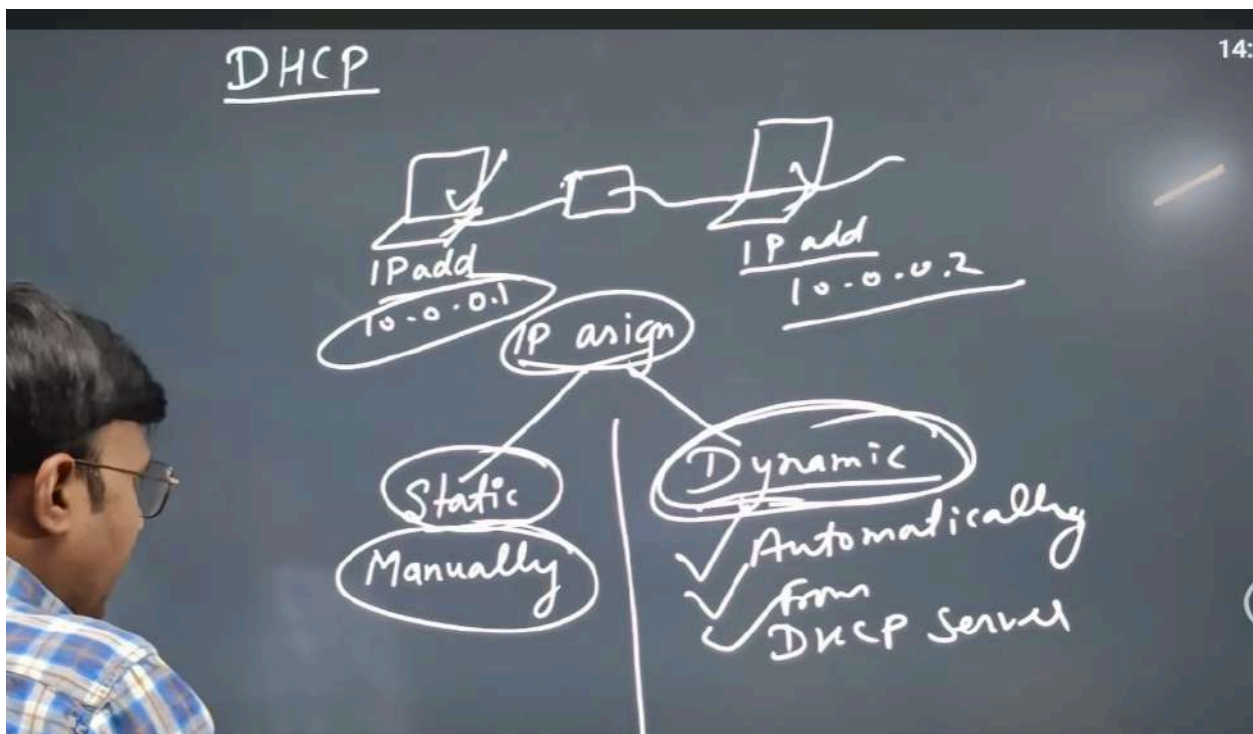
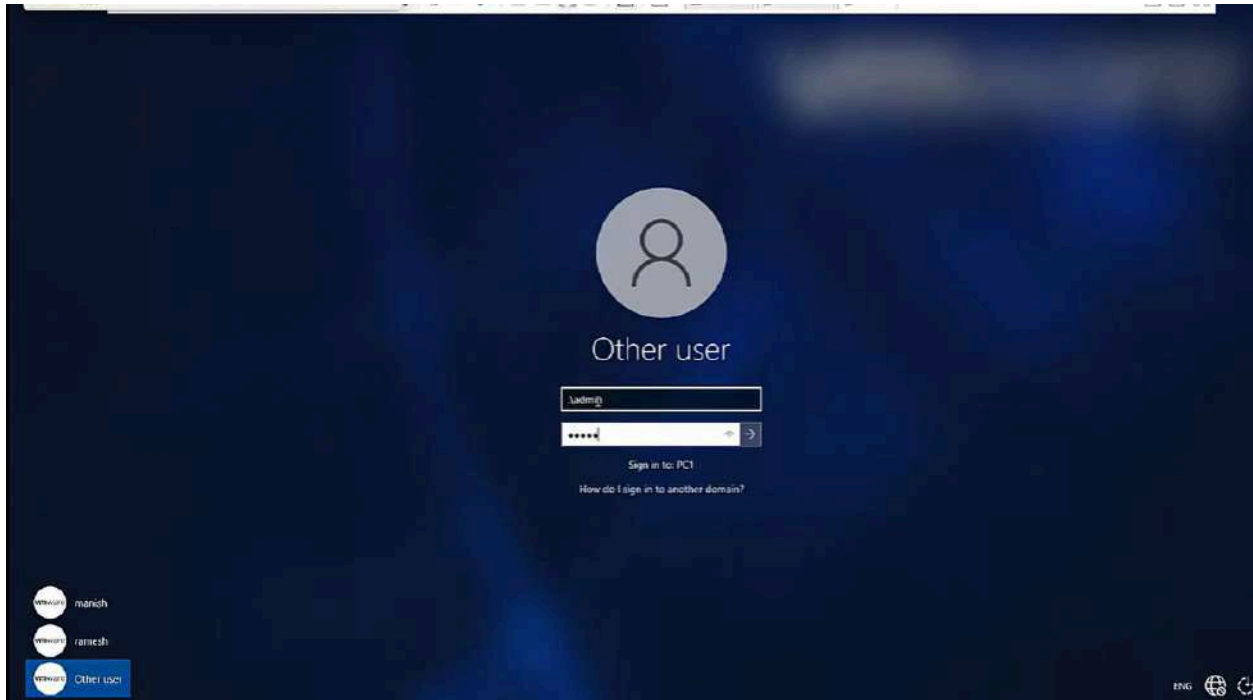
Create Bulk user at a time(command):





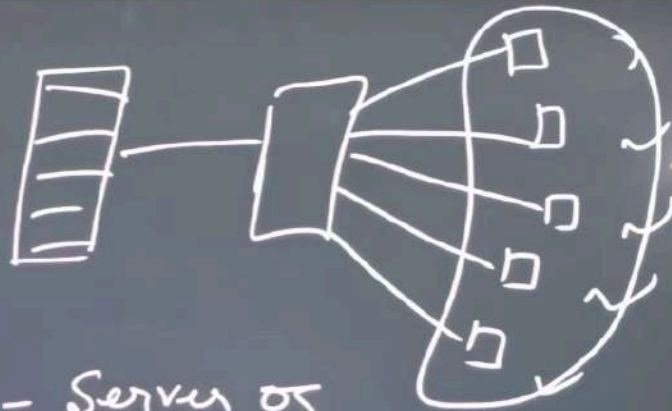
- Active Directory Users and Computers
 - Saved Queries
 - icntacademy.com
 - Builtin
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - Managed Service Accounts
 - Users
 - ICNTGROUP
 - ICNT ACADEMY
 - ICNT GYAN
 - ICNT CLOUD GYAN

Name	Type	Description
user16	User	
user17	User	
user18	User	
user19	User	
user2	User	
user20	User	
user3	User	



DHCP

15:02



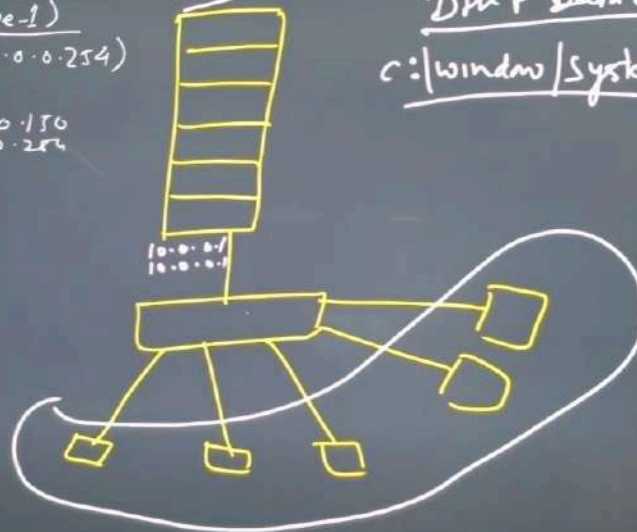
- Server OS
- Router
- Layer 3 Switches

DHCP.

15:10

Scope name (Scope-1)
IP Range (10.0.0.2 - 10.0.0.254)
Excluded IP Range
10.0.0.101 - 10.0.0.150
10.0.0.101 - 10.0.0.254

Win 2022



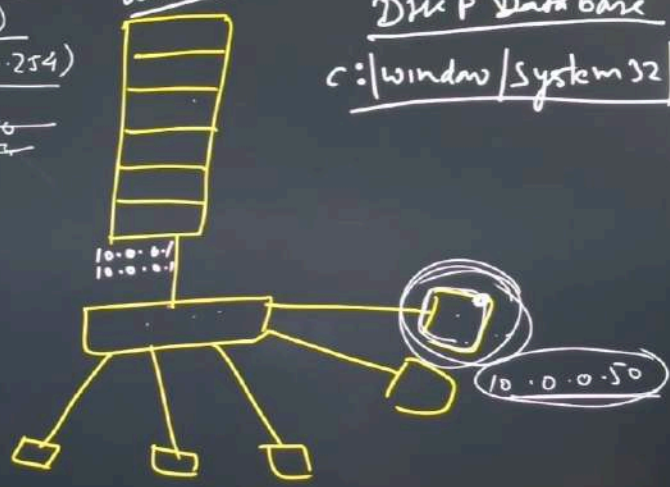
DHCP Database
c:\windows\system32\dhcp

DHCP.

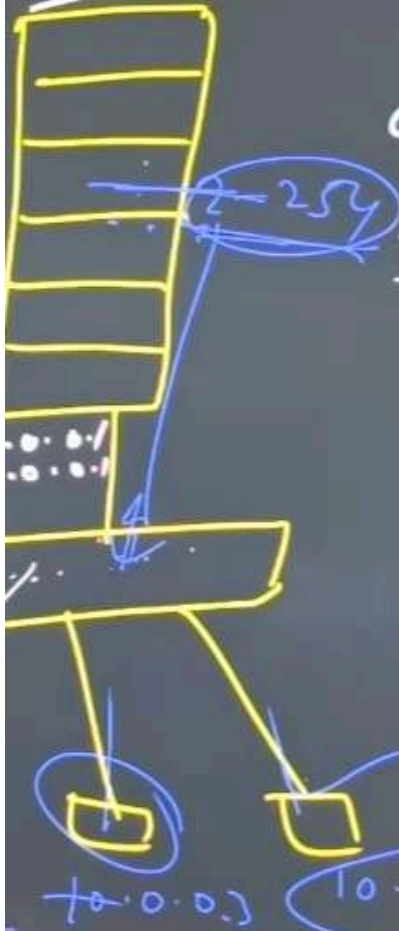
Scope name (Scope-1)
 IP Range (10.0.0.2-10.0.0.254)
 Excluded IP Range
~~10.0.0.101-10.0.0.150~~
~~10.0.0.101-10.0.0.254~~
 Default Gateway
 10.0.0.100
 Reserved IP Range
 10.0.0.50

Win 2022

DHCP Database
 c:\windows\system32\dhcp



Win 2022



DHCP Database

c:\windows\system32\dhcp

DHCP Lease Process

D → Discover — by client

O — offer → by server

R — Request → client

A — Acknowledgement → server

Lease Duration

min — 1 min

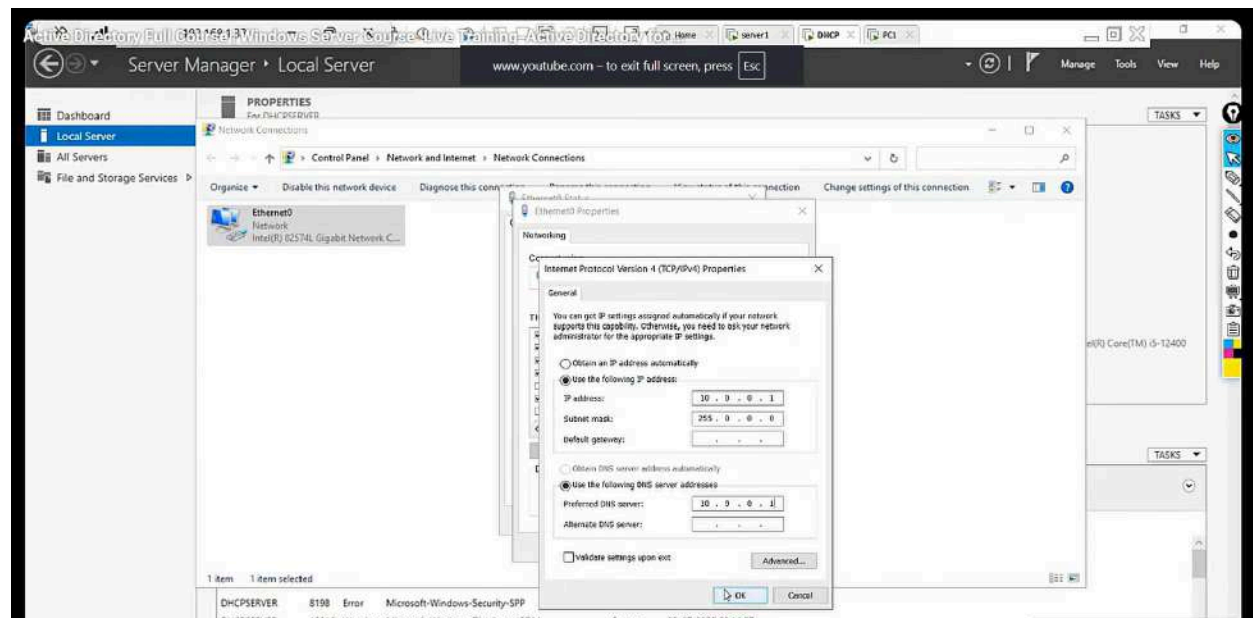
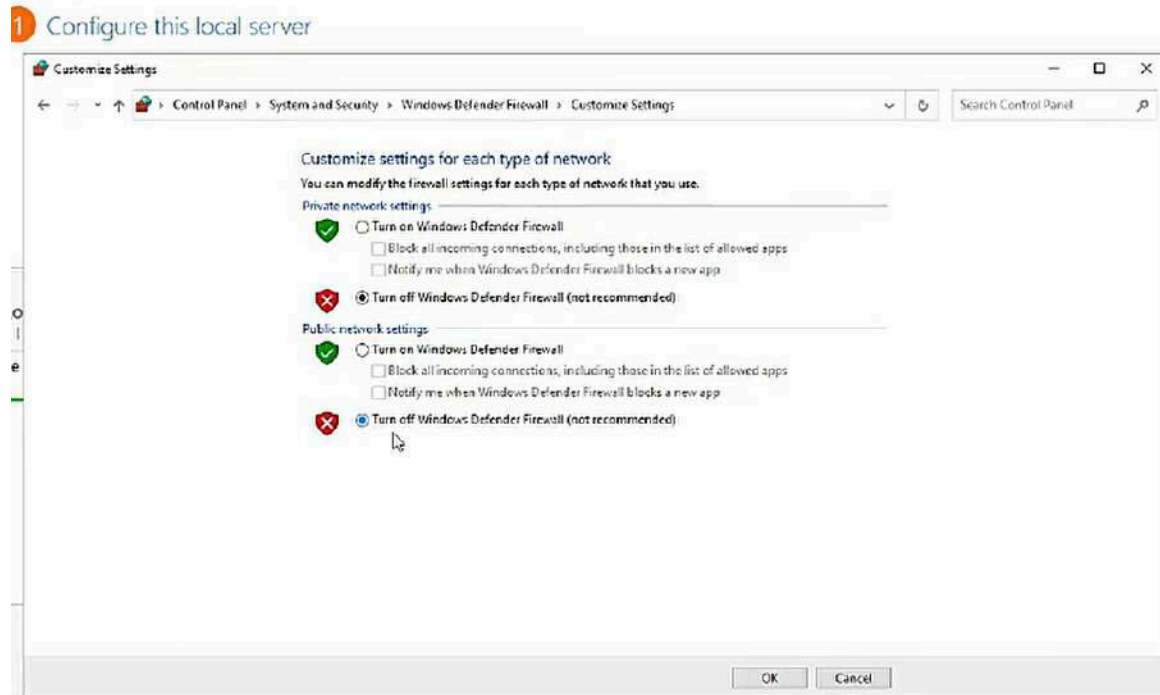
by default — 8 days

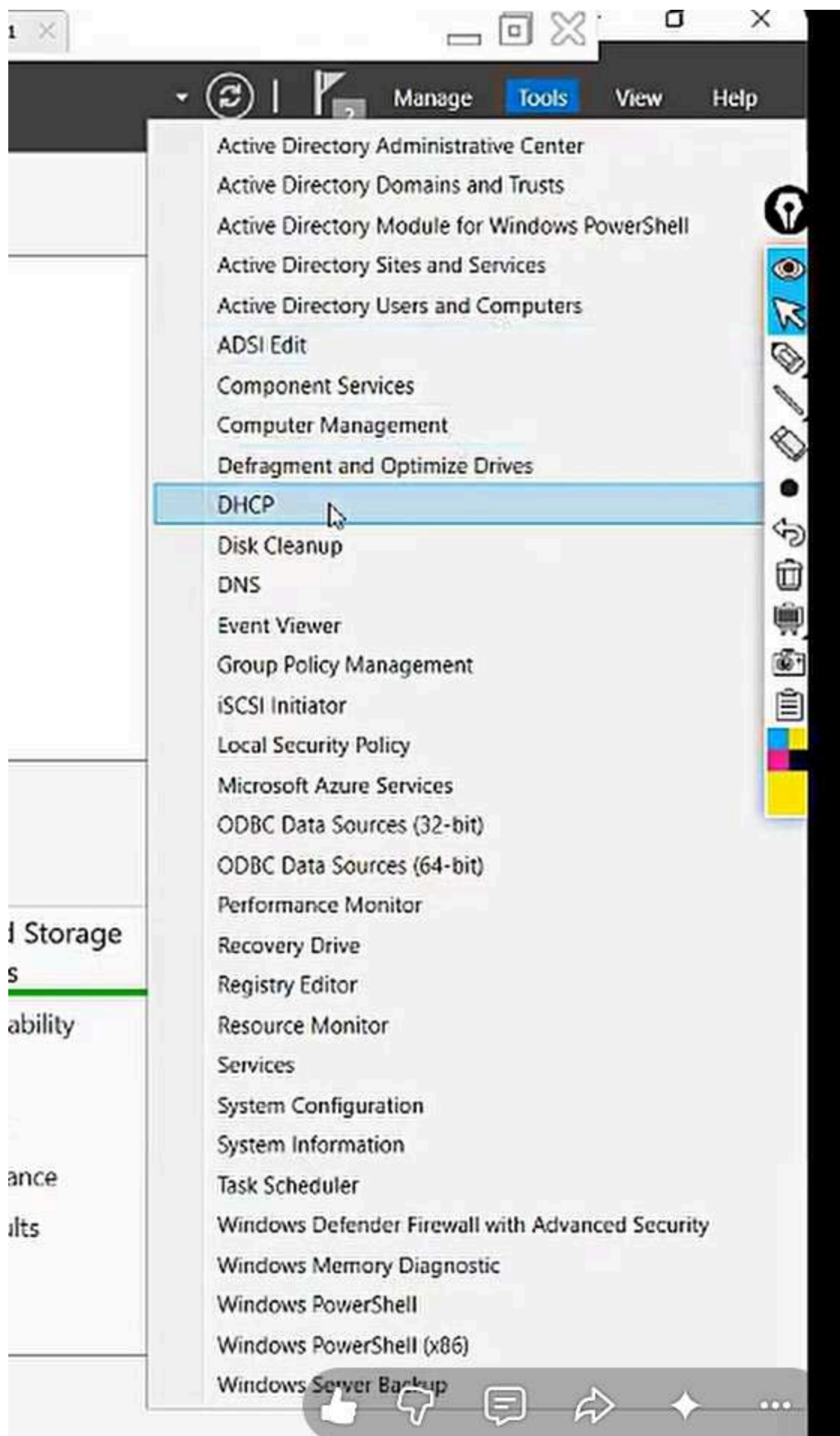
max — 999 days 23hrs 59min

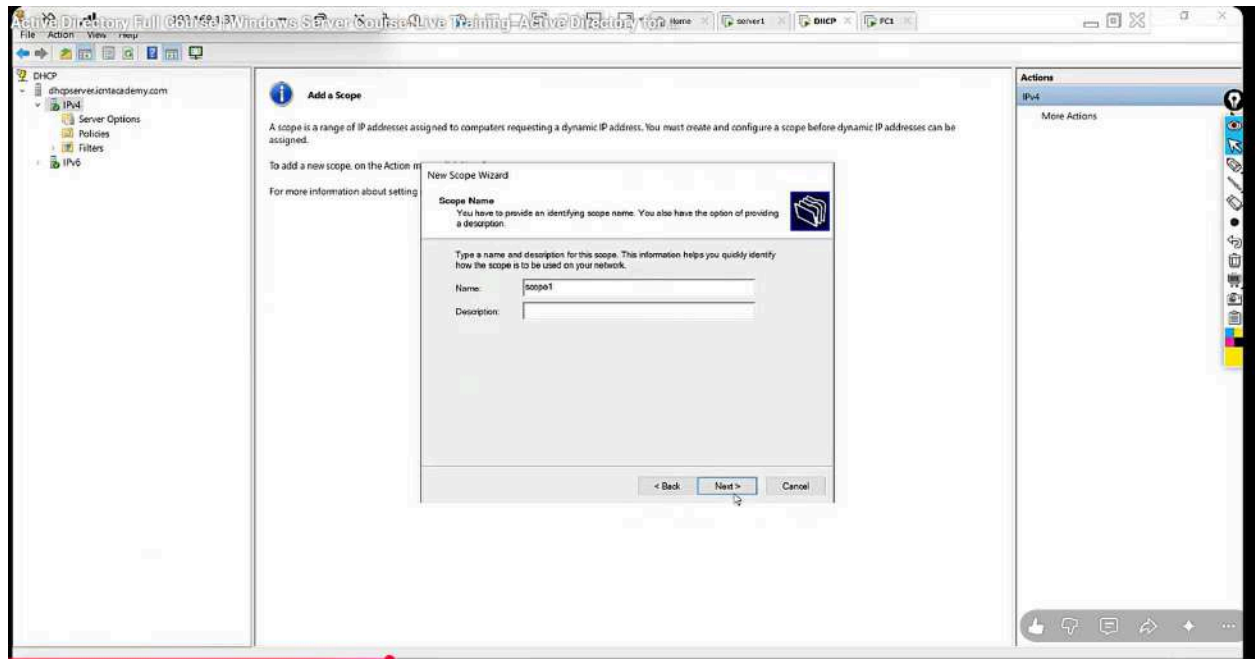
DHCP Sever Setup:

Before doing something we have to turn off Firewall services

And provide to static ip address to DHCP Server







IP addresses assigned to computers requesting a dynamic IP address. You must create and configure a scope before dynamic IP addresses

on the Action menu
n about setting

New Scope Wizard

IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 10 . 0 . 0 . 2

End IP address: 10 . 0 . 0 . 254

Configuration settings that propagate to DHCP Client

Length: 8

Subnet mask: 255 . 0 . 0 . 0

< Back Next > Cancel

ction m

setting

New Scope Wizard

Add Exclusions and Delay

Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.

Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

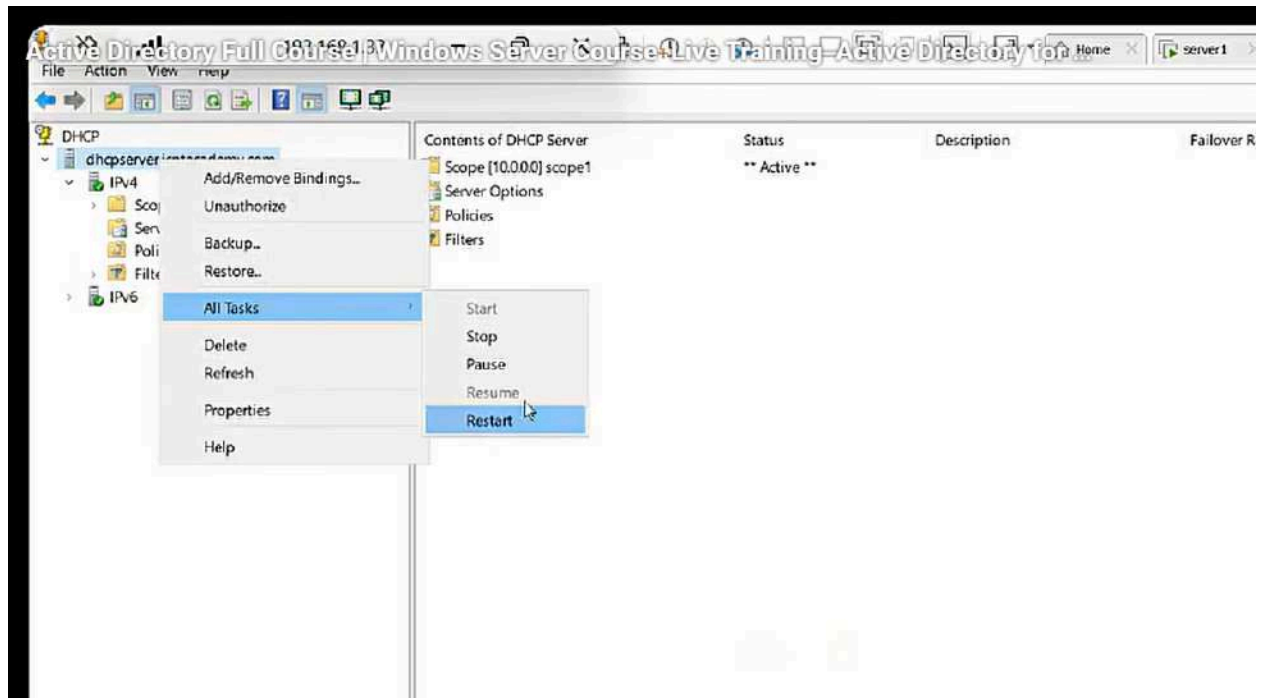
Start IP address: End IP address:

Excluded address range:

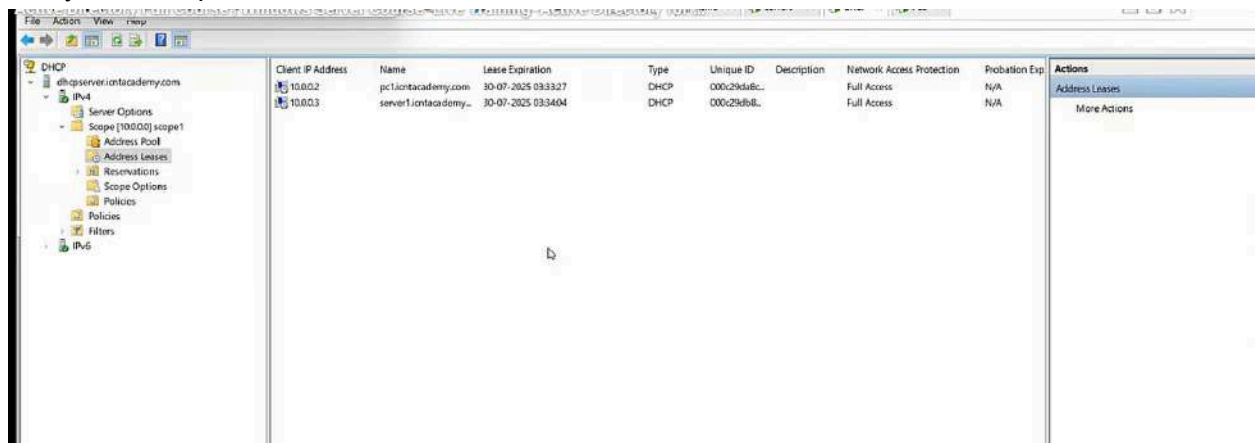
10.0.0.150 to 10.0.0.200	Remove
--------------------------	---------------------------------------

Subnet delay in milli second:

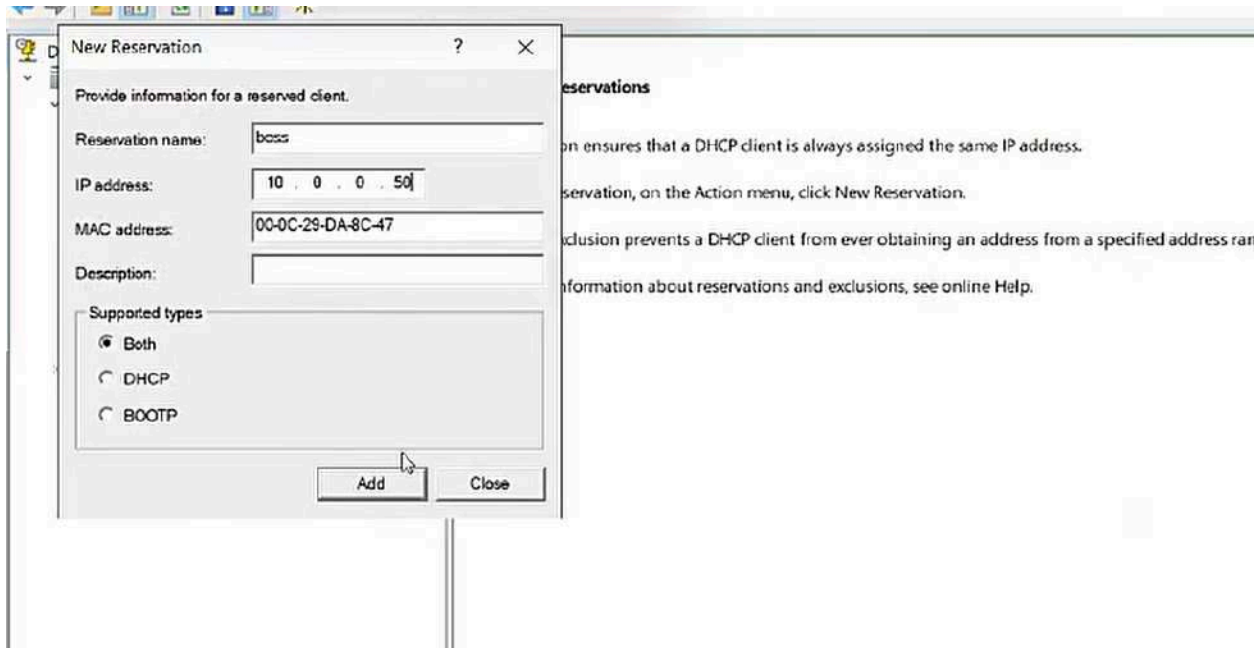
For activate we have to restart the DCP



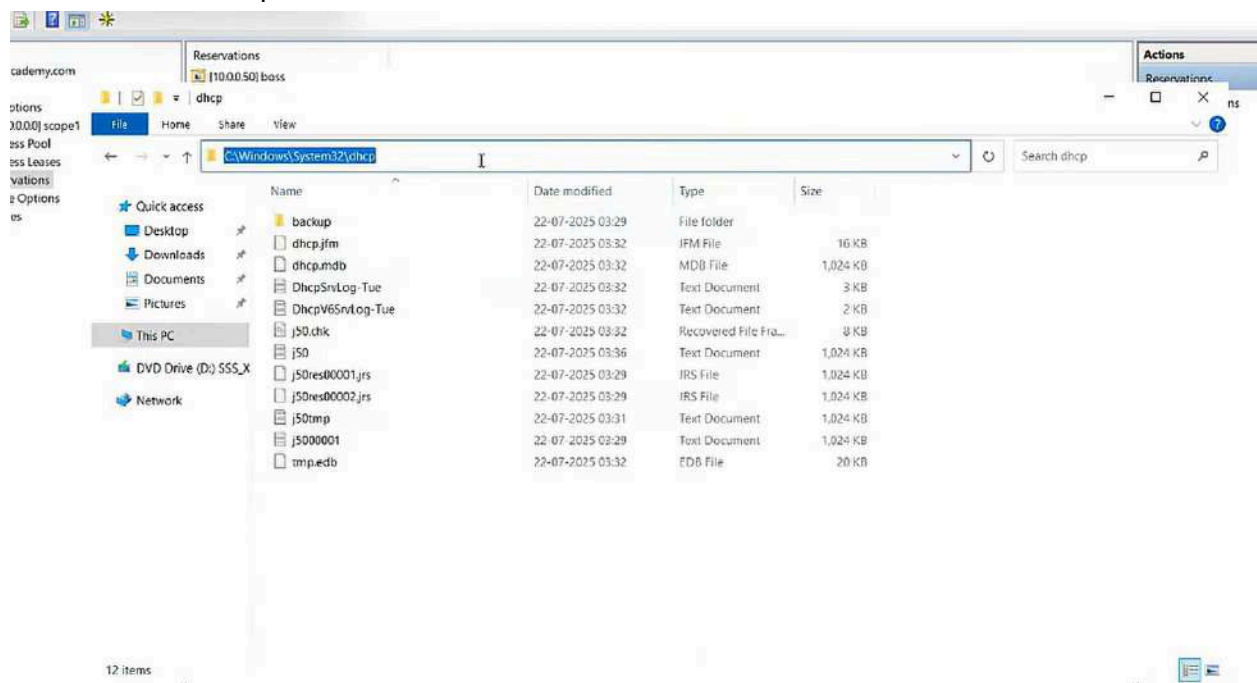
Everyone Get ip:

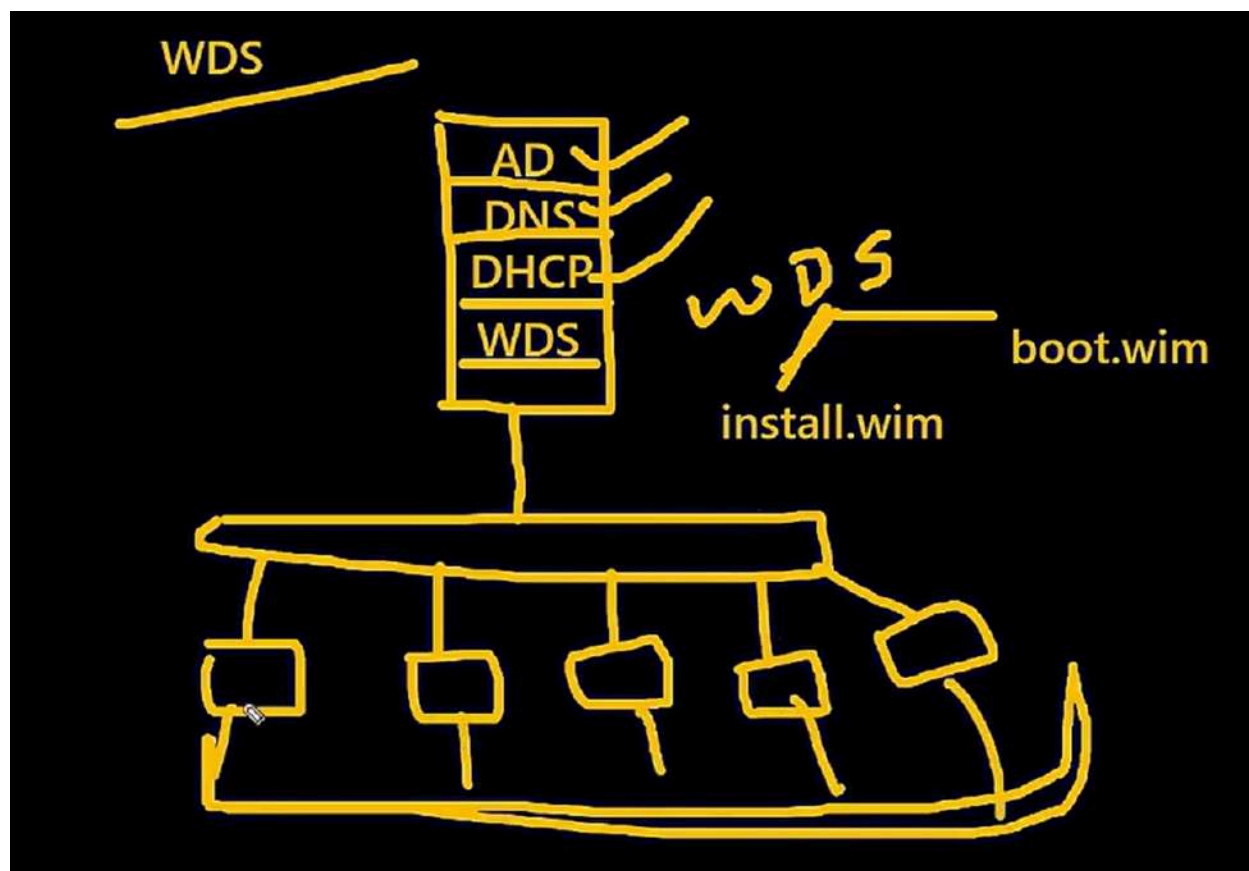


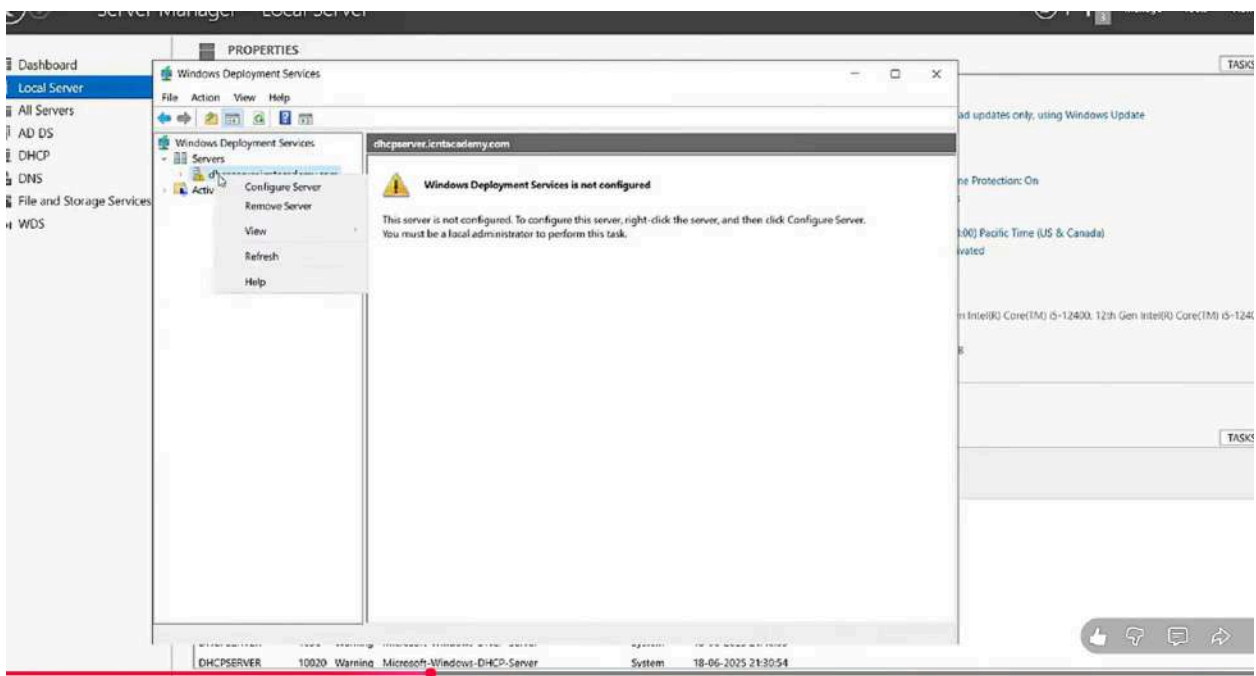
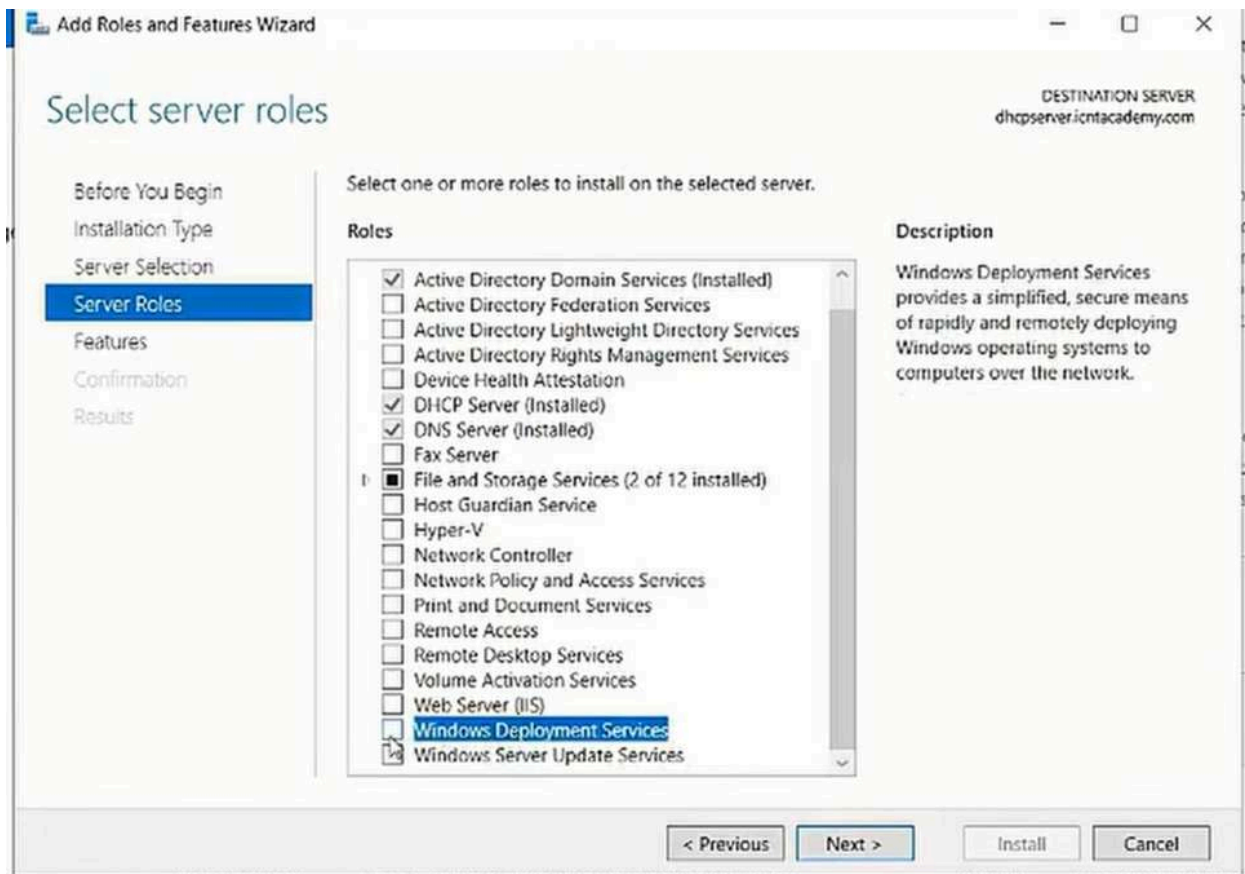
For who need specific Ip address :

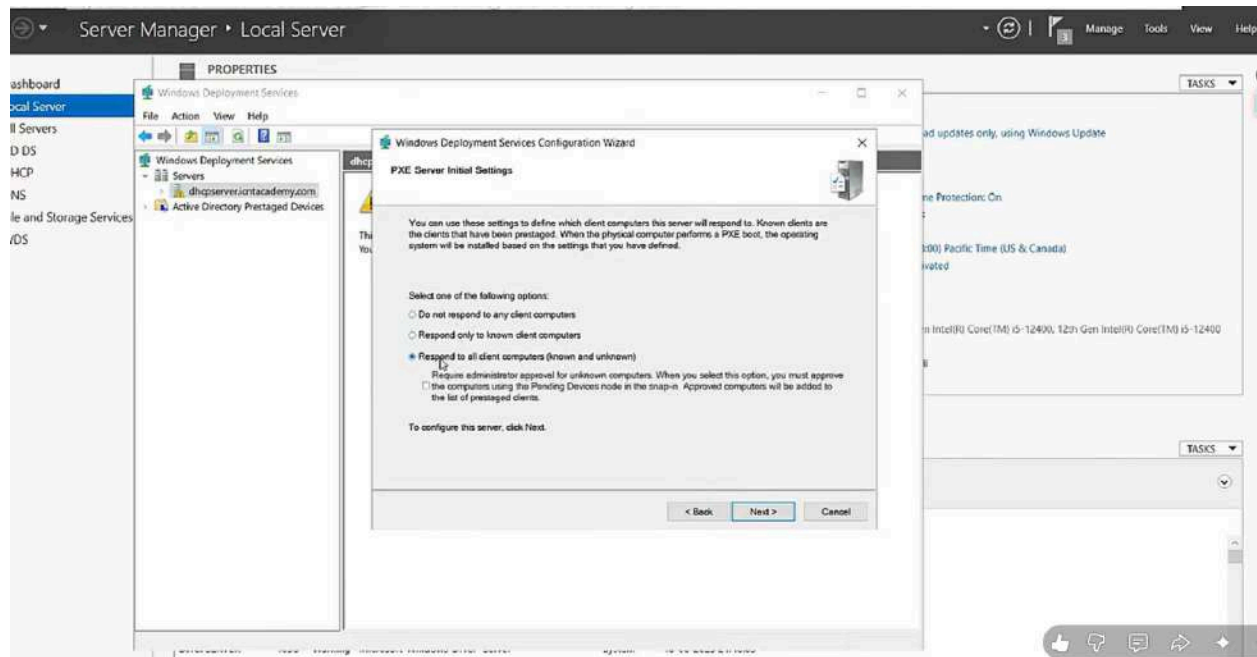


If we want to Backup DHCP:

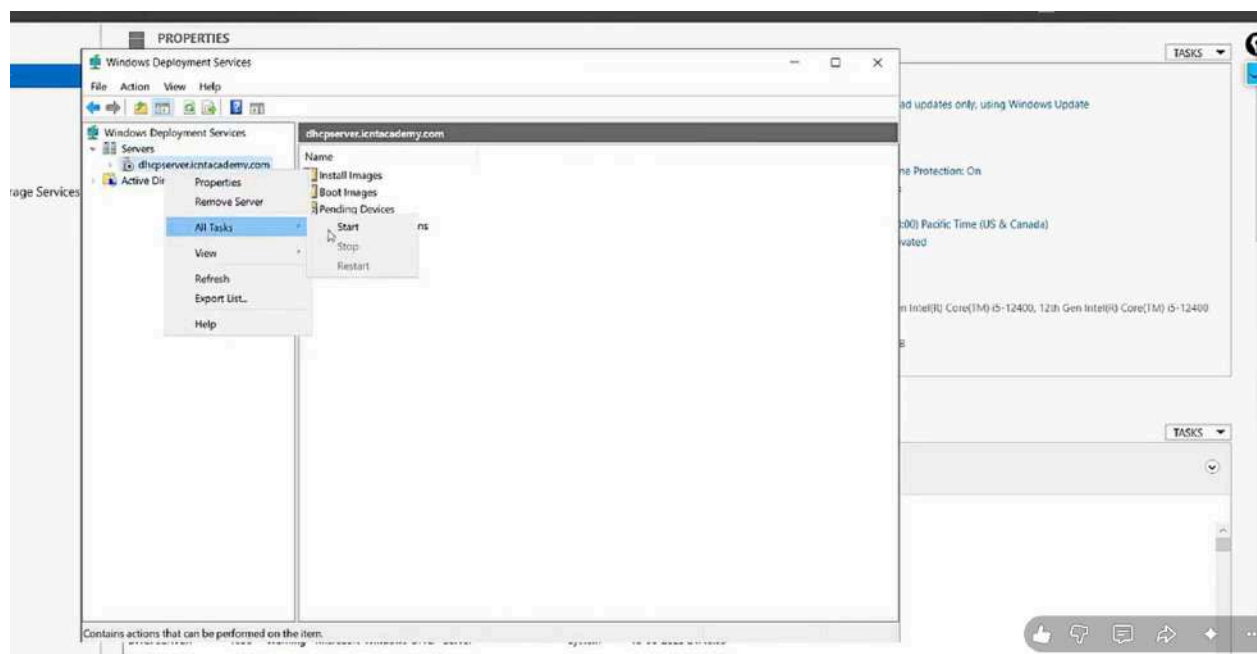




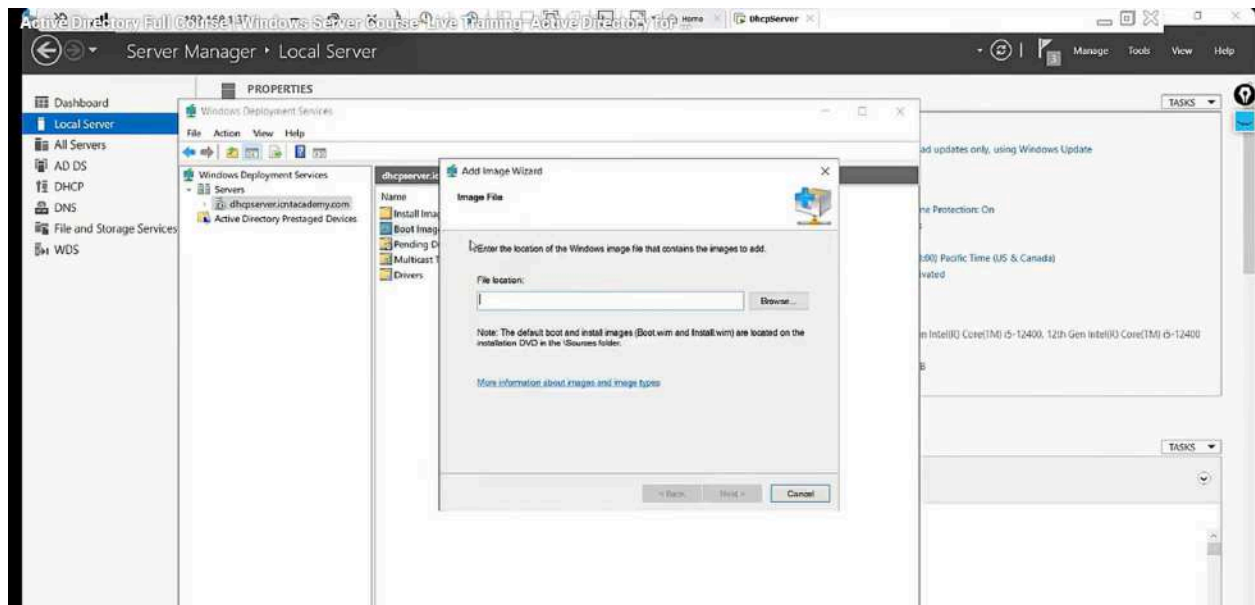




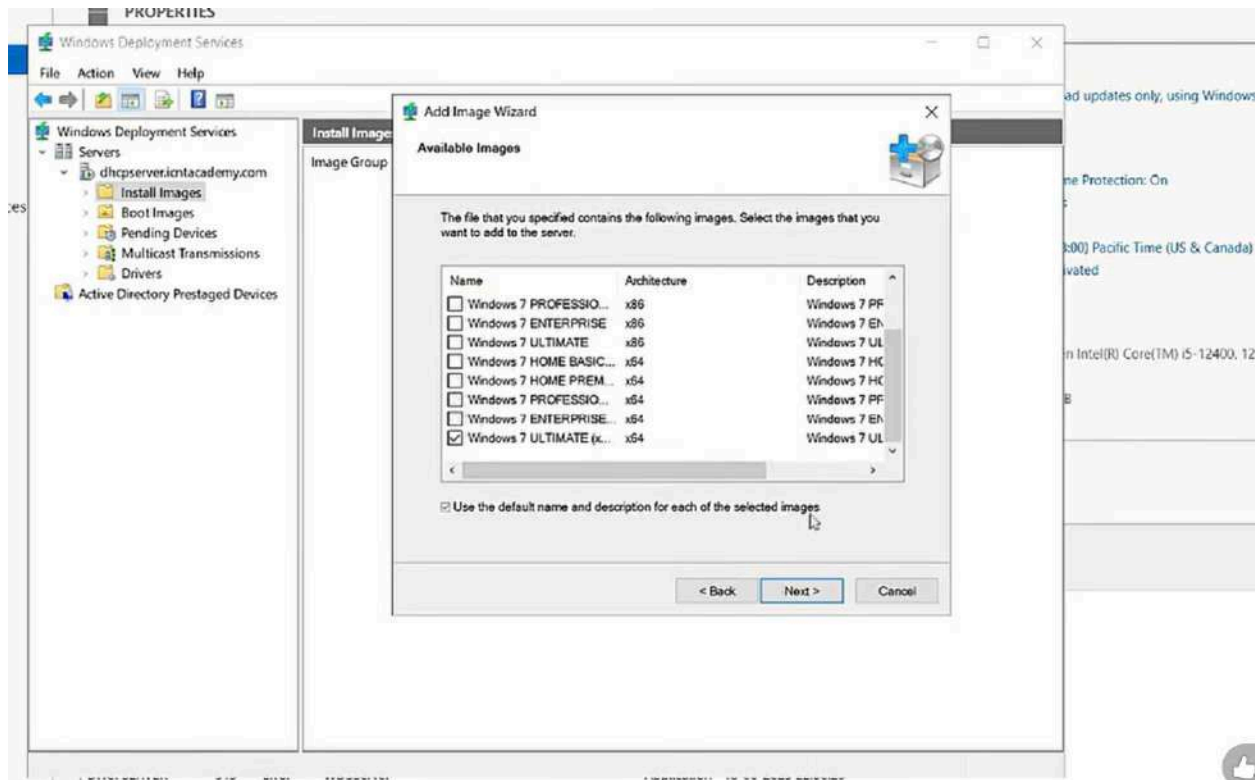
For Restart WDS:



Add Boot image:
Select boot.wim



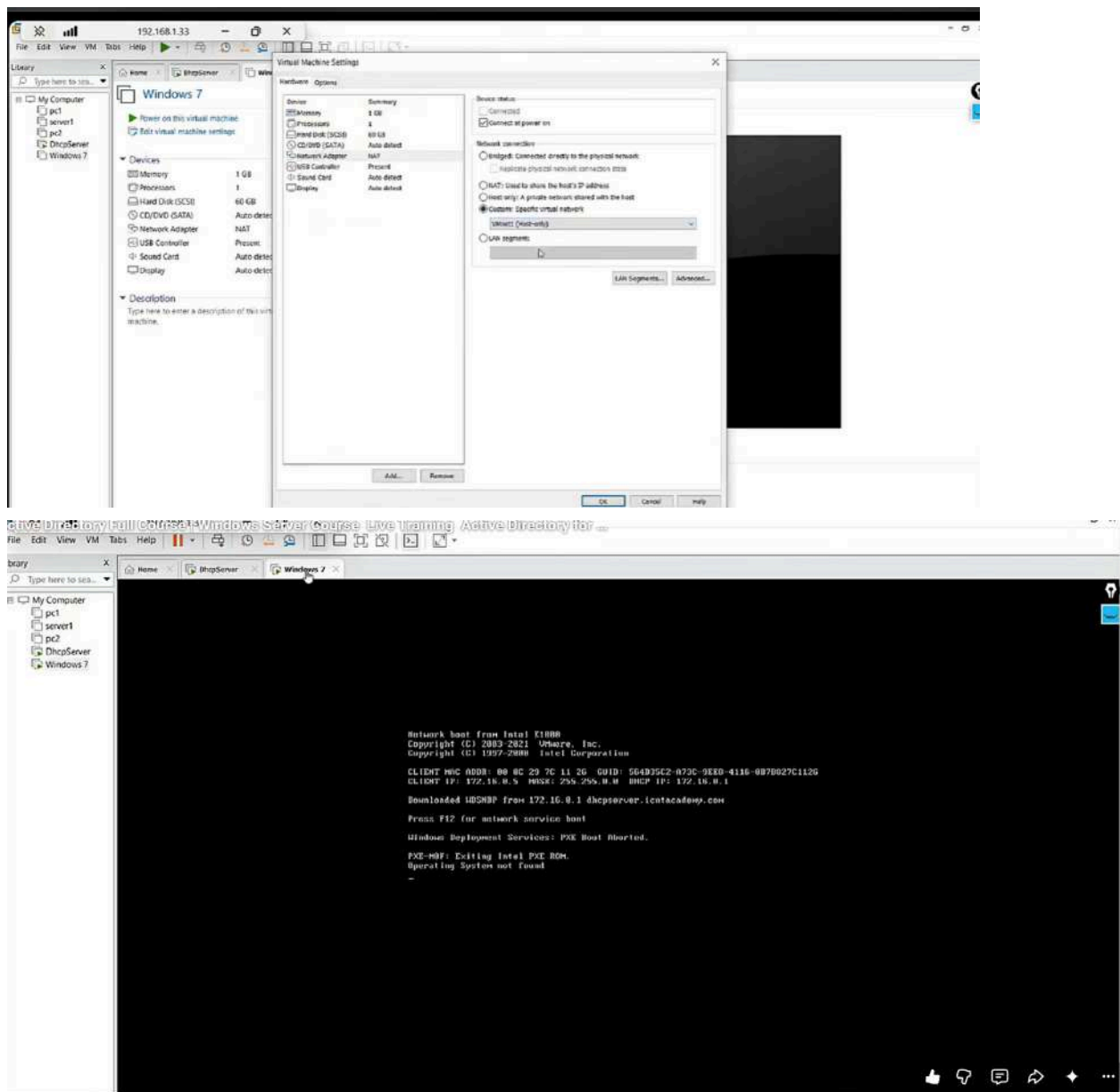
And also add install.wim in install images:



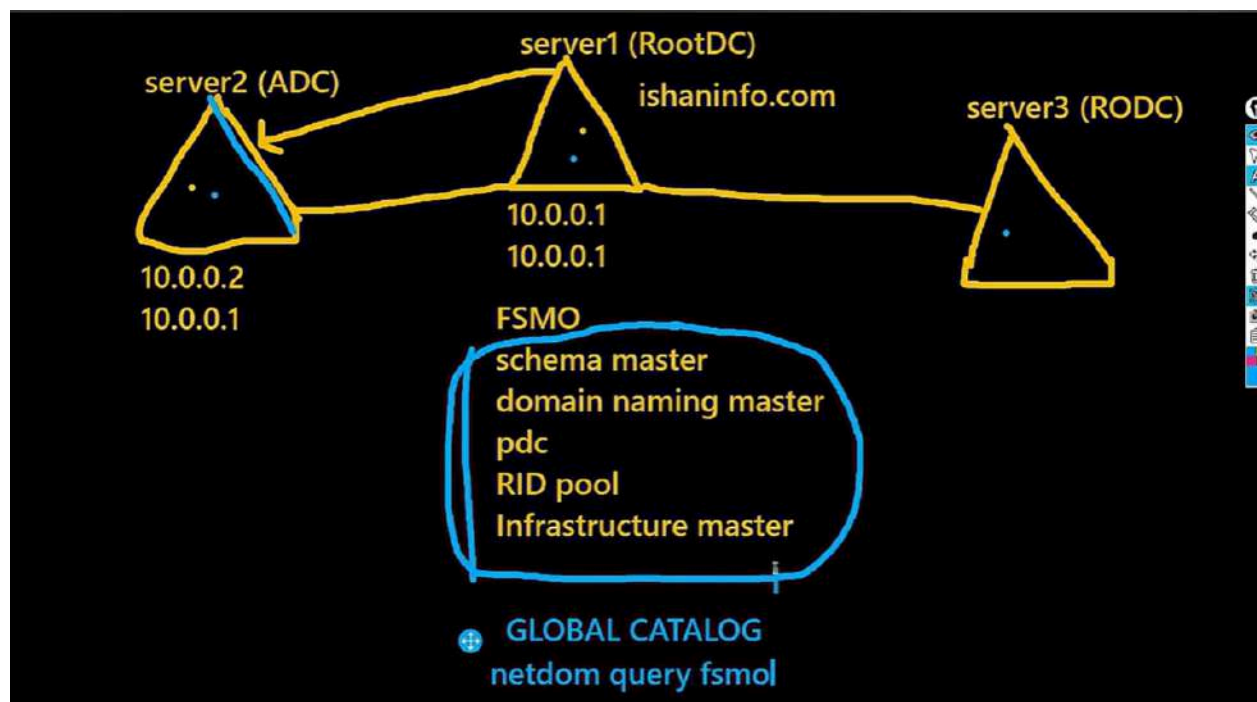
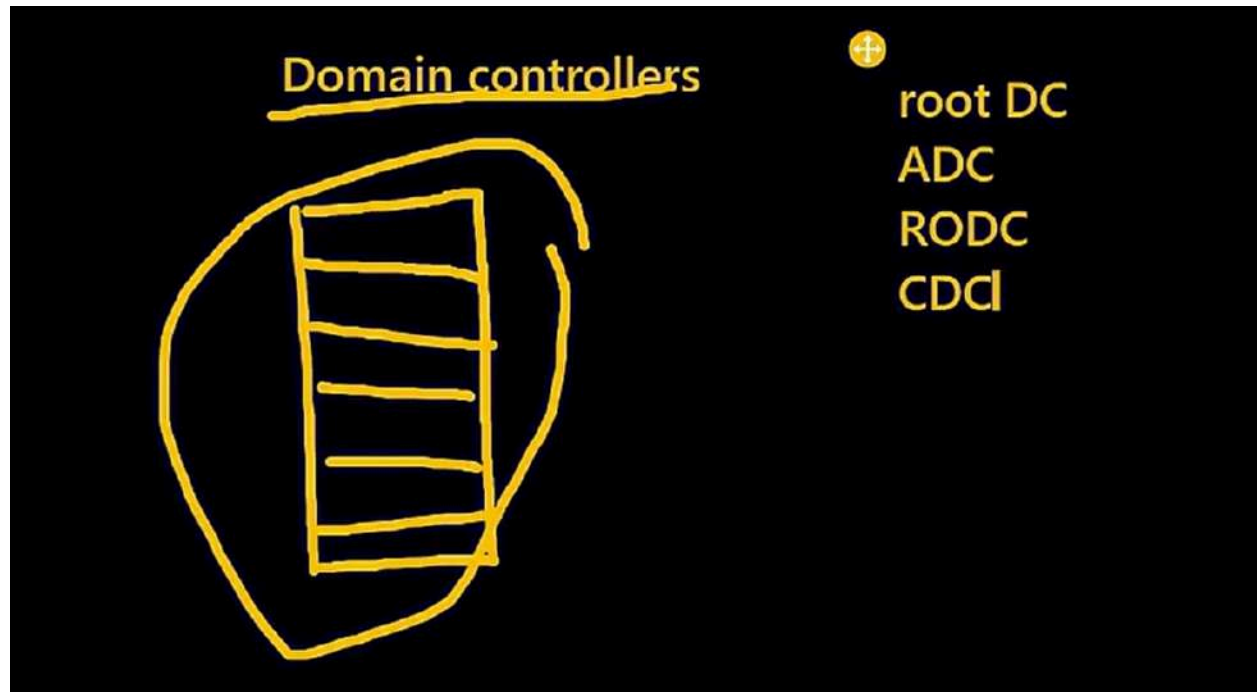
After the WDS server is configured, client PCs can install the operating system directly from the server. Simply start the client PC, press **F12** to boot from the network (PXE), and the installation will begin automatically.

Testing a New Client PC

Create a new virtual machine without attaching an ISO file. When the VM starts, immediately press **F12** to perform a PXE boot and obtain an IP address from the DHCP server. The operating system installation will then start from the WDS server.

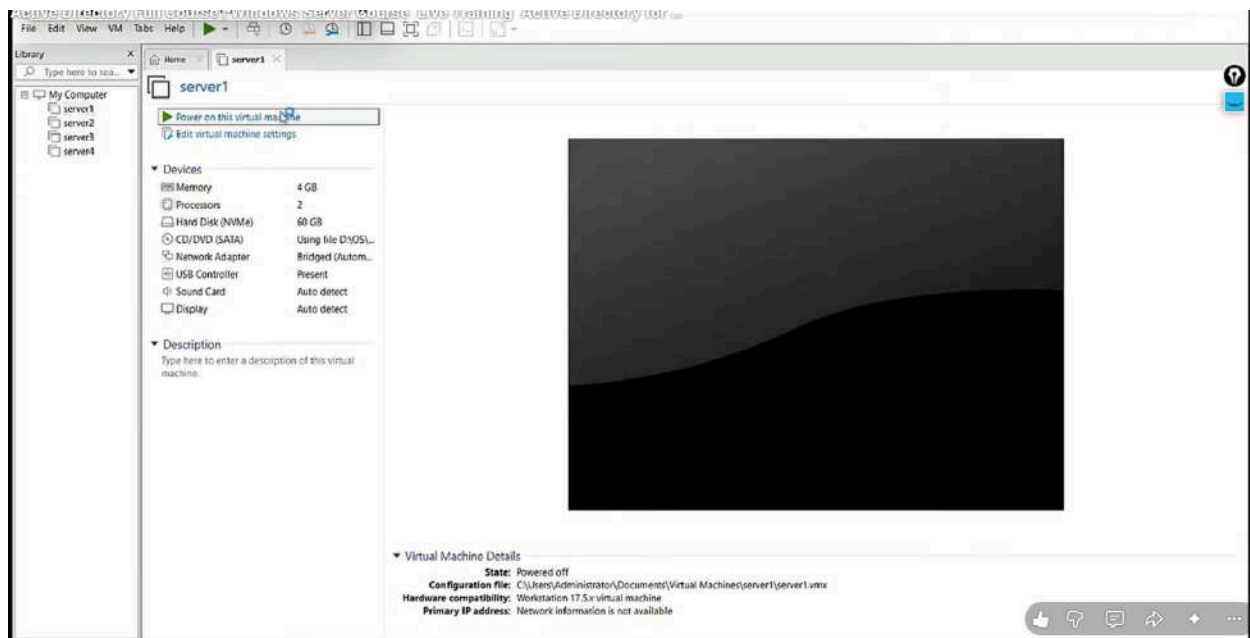


Domain Controller:

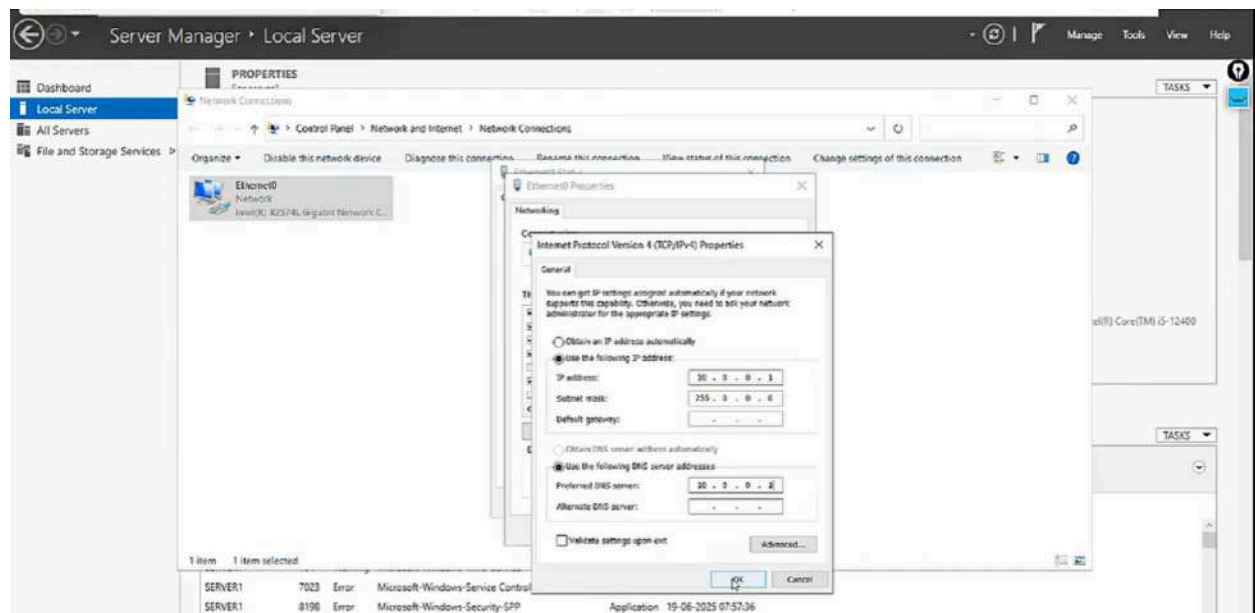


Create four servers to demonstrate the roles of different Domain Controllers:

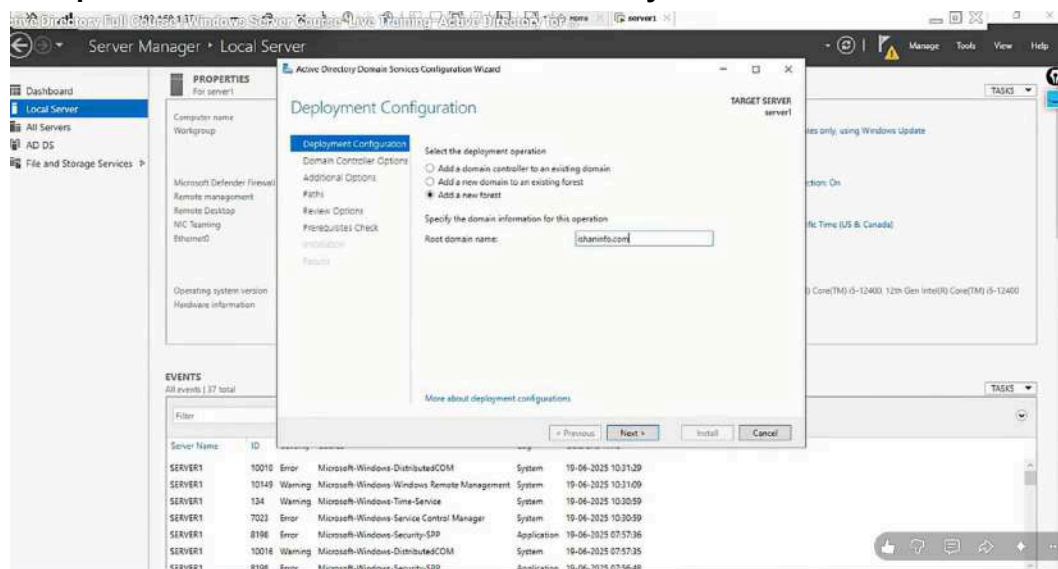
- **Server1 – Root Domain Controller (Root DC)**
- **Server2 – Additional Domain Controller (ADC)**
- **Server3 – Read-Only Domain Controller (RODC)**
- **Server4 – Child Domain Controller (CDC)**



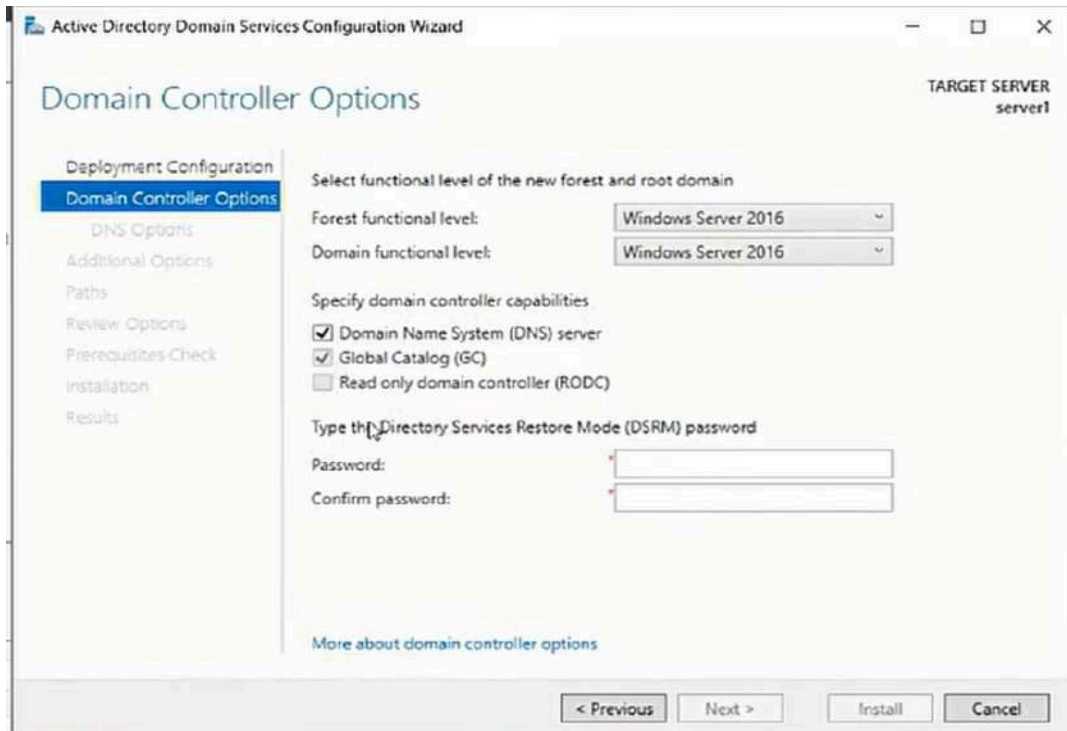
Now, we want to configure Server1 as the Root Domain Controller (RDC). Before installing Active Directory Domain Services (AD DS), assign a static IP address because a Domain Controller requires a fixed IP address.



Now promote the server as Root DC or Primary DC:



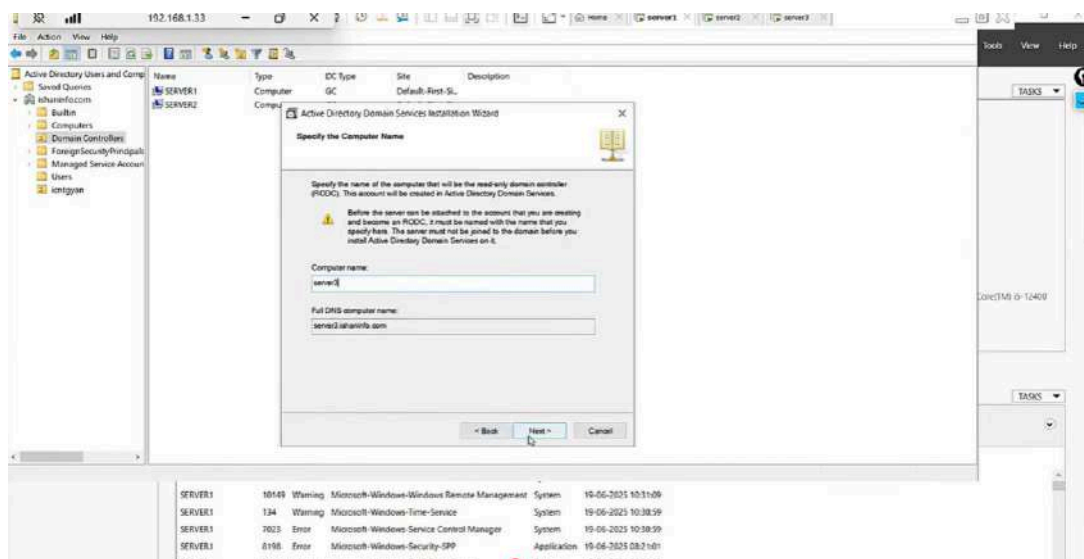
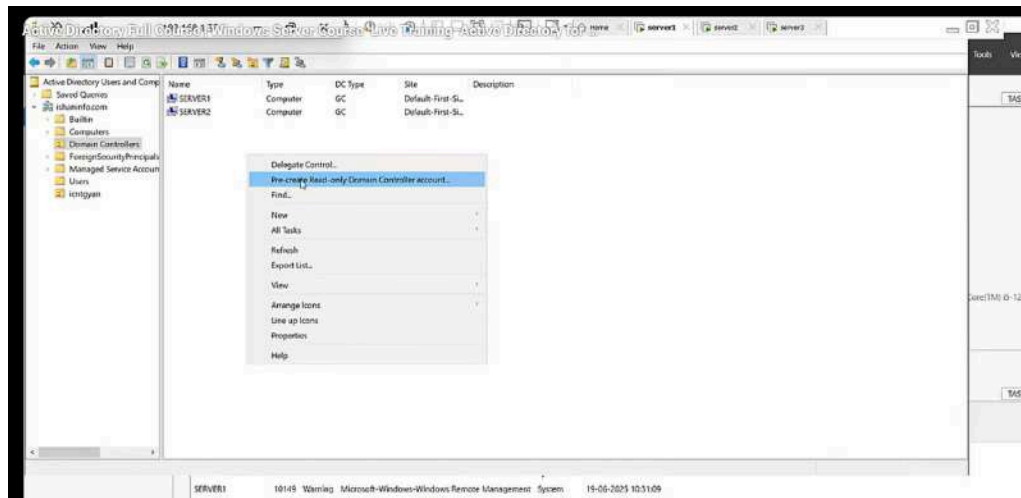
When we create a new forest or promote a server as the Root Domain Controller, we cannot remove the Global Catalog (GC) from that Domain Controller.



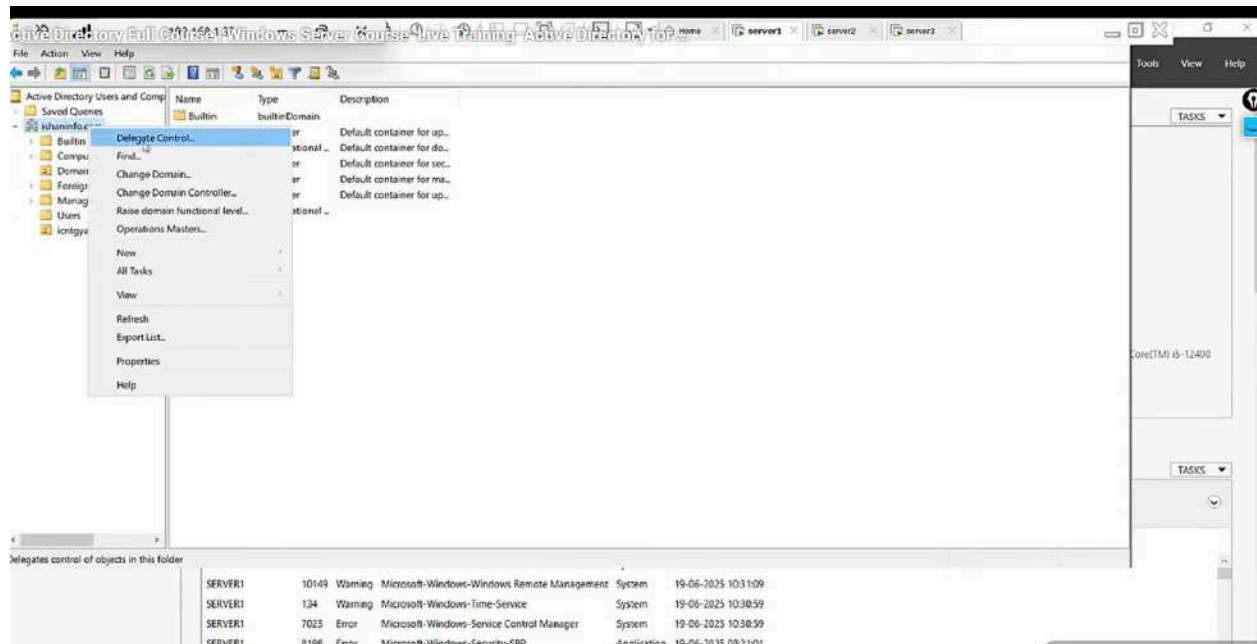
In 5:13:00minute I have doubt on this topic



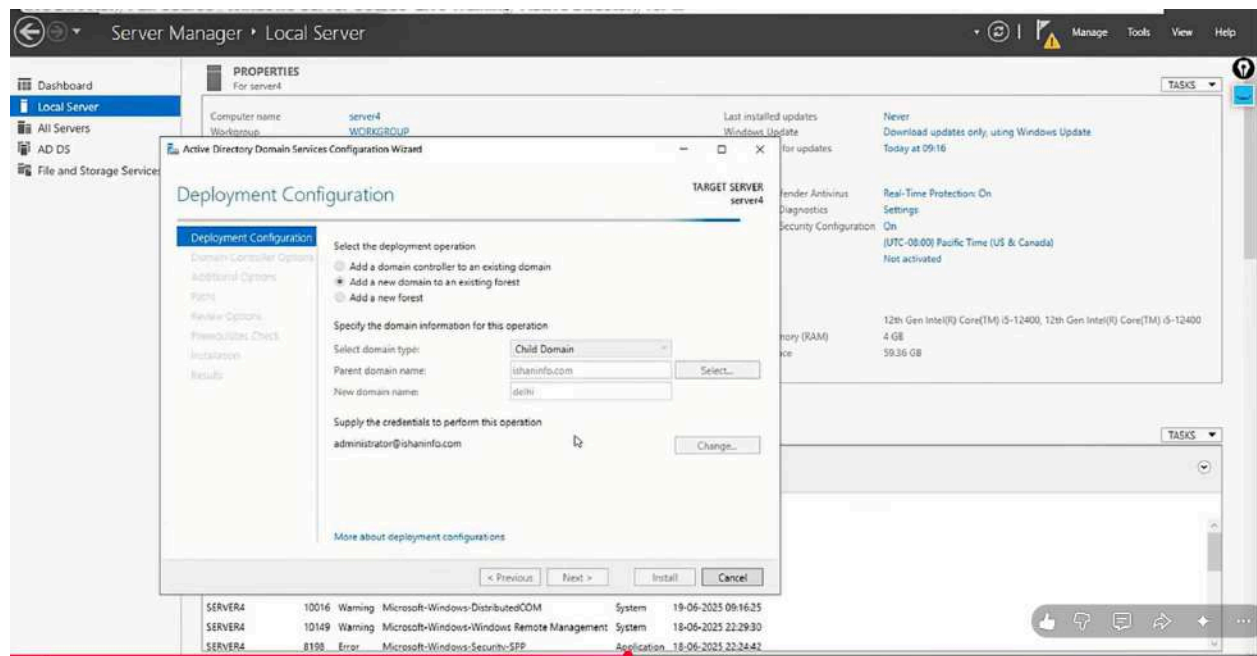
Server3 is the Read-Only Domain Controller (RODC). To configure its delegated administration, we need to perform the setup on Server1. In Active Directory Users and Computers, select Domain Controllers.



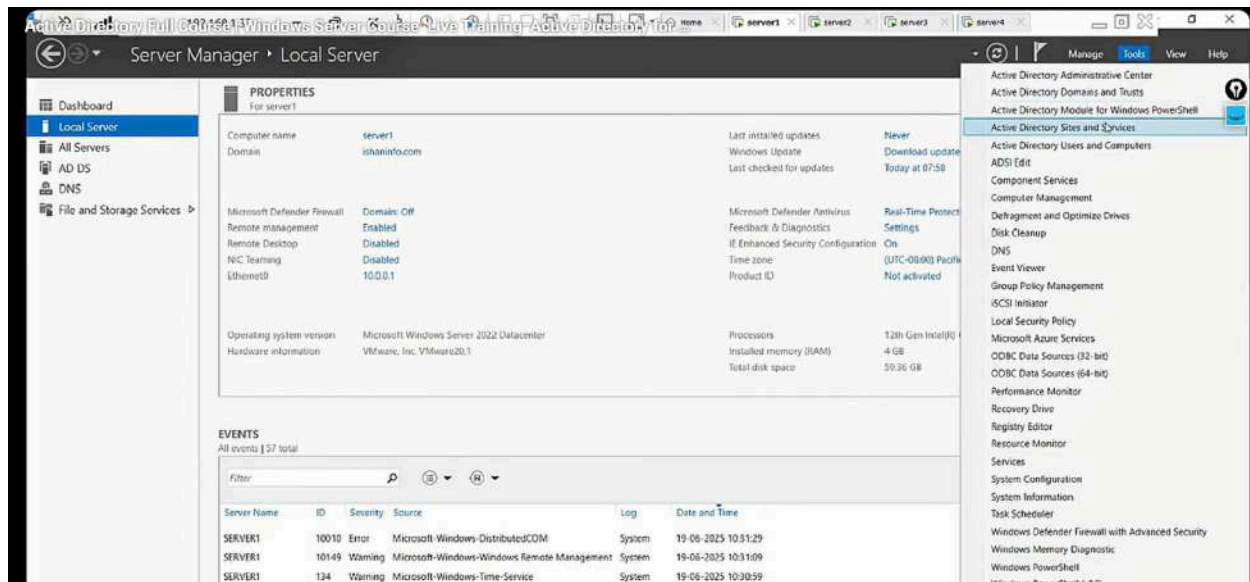
Delegate Control means assigning specific administrative tasks to a Read-Only Domain Controller (RODC). In this example, Server3 is the RODC.



Setup Child Domain(CDC):

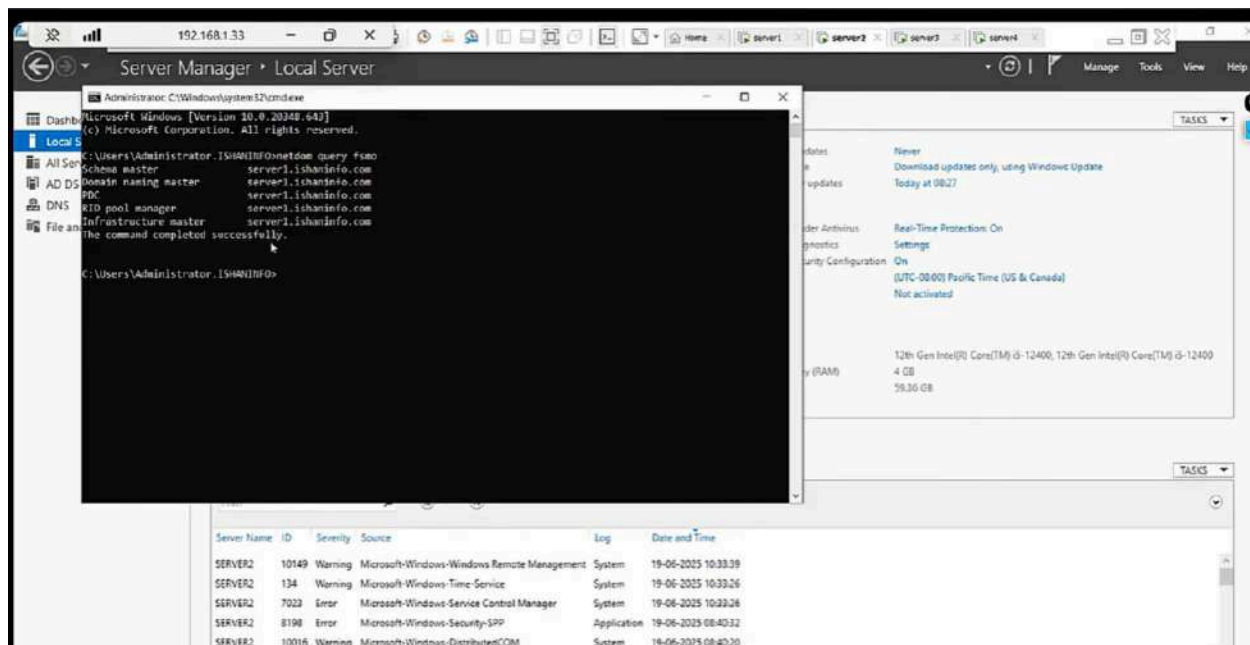


To see Child Domain from Root DC:



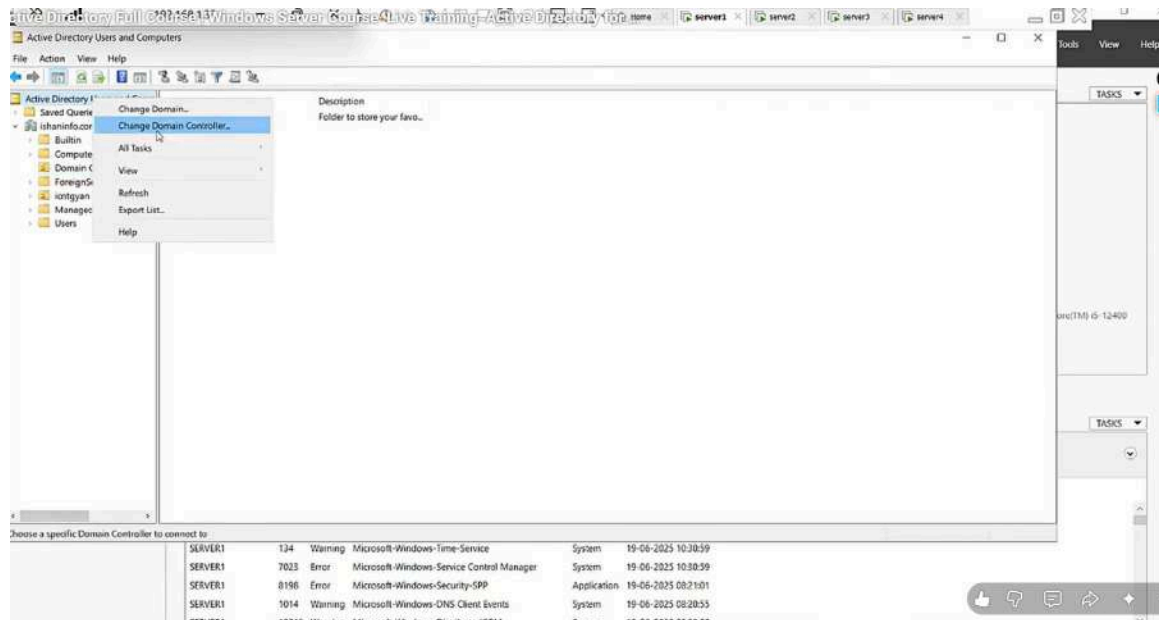
Role Transfer to Server2

Currently, all roles are hosted on Server1.

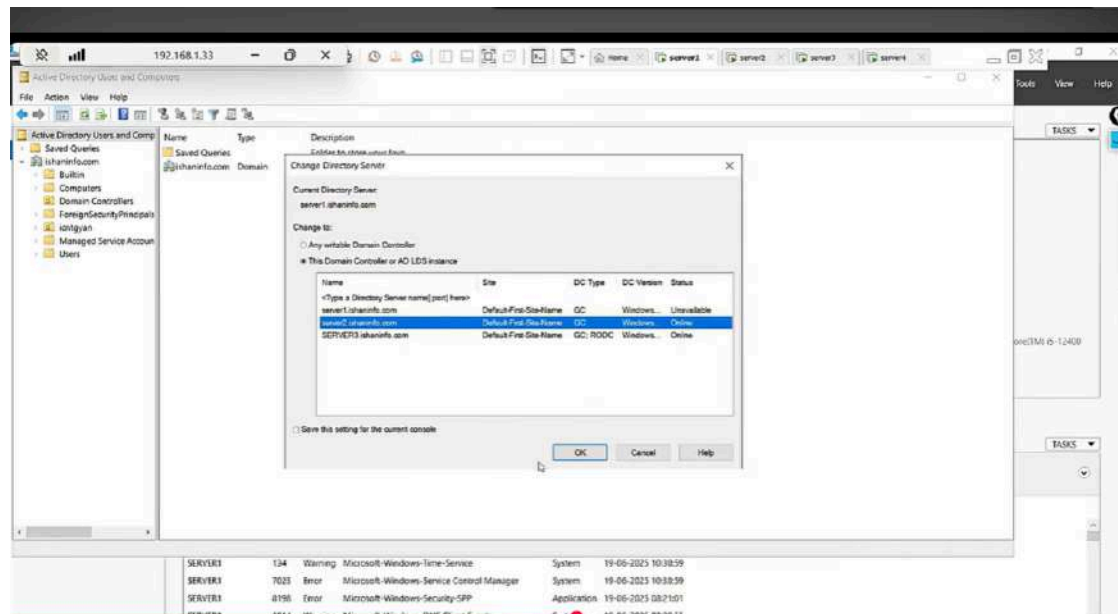


Now, we are going to transfer all the roles to Server2 so that it becomes our primary Root Domain Controller.

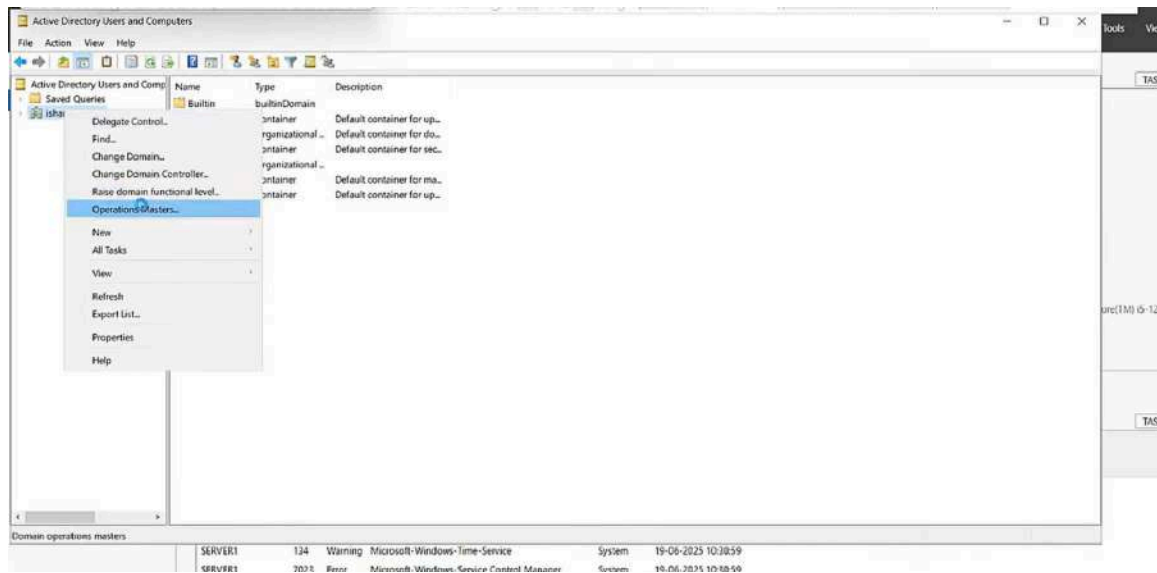
First Select change Domain Controller from server1



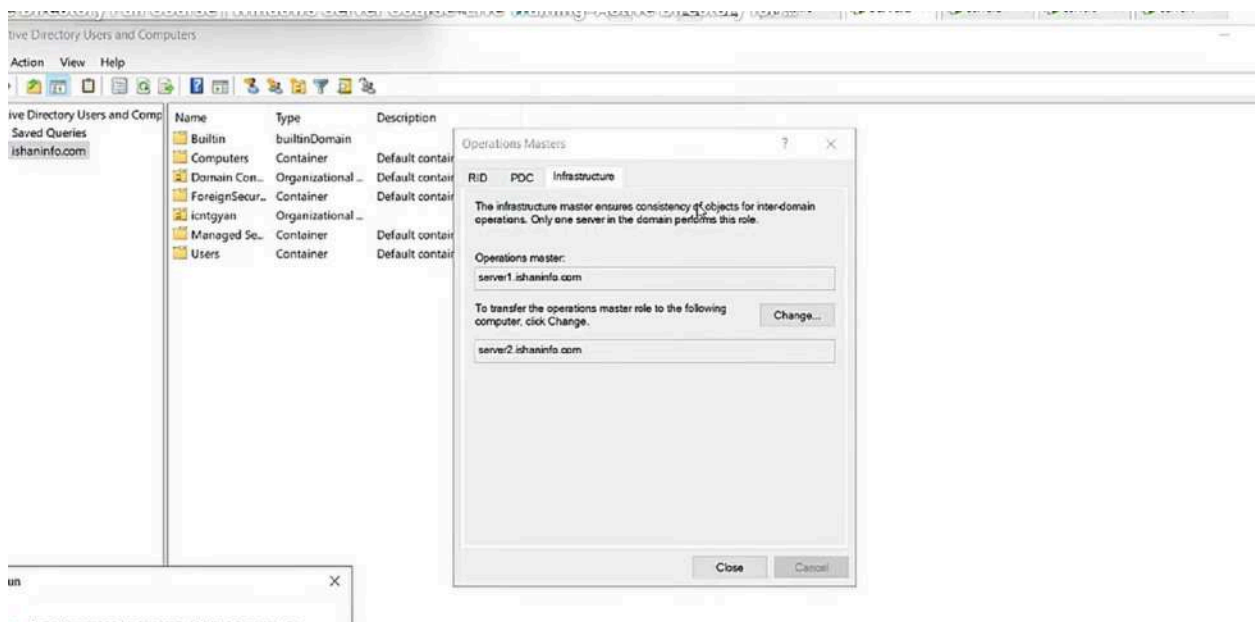
Then select server2



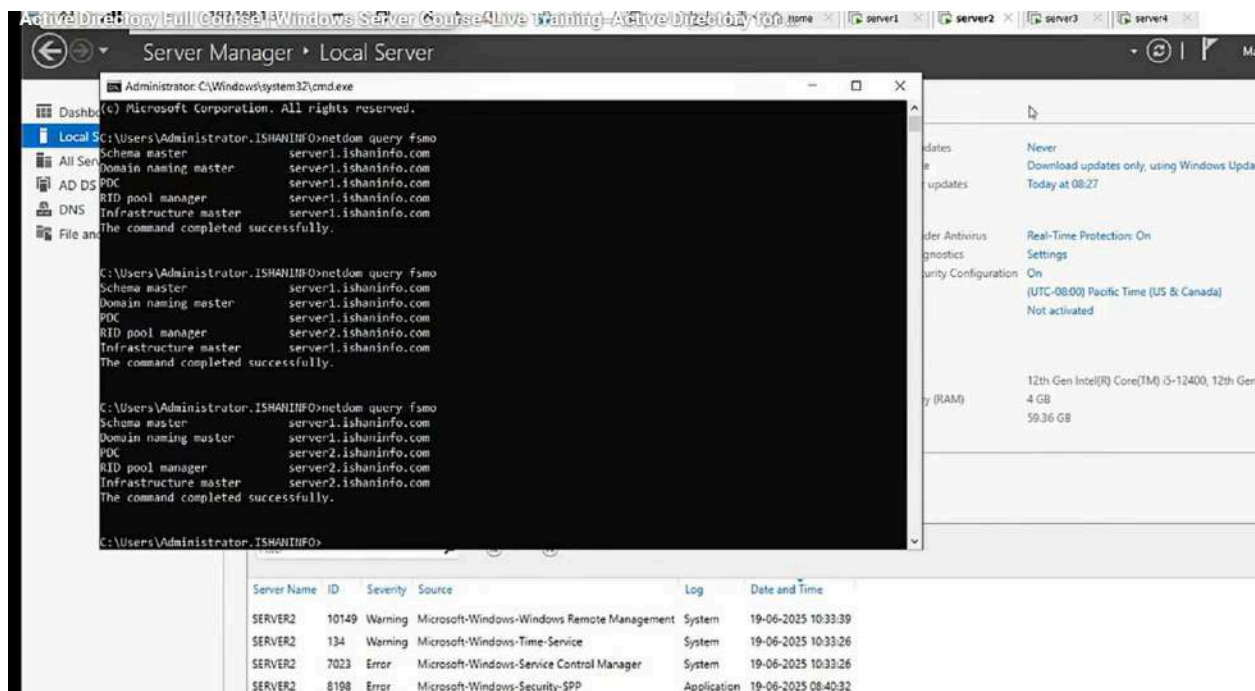
Then go to operation master:



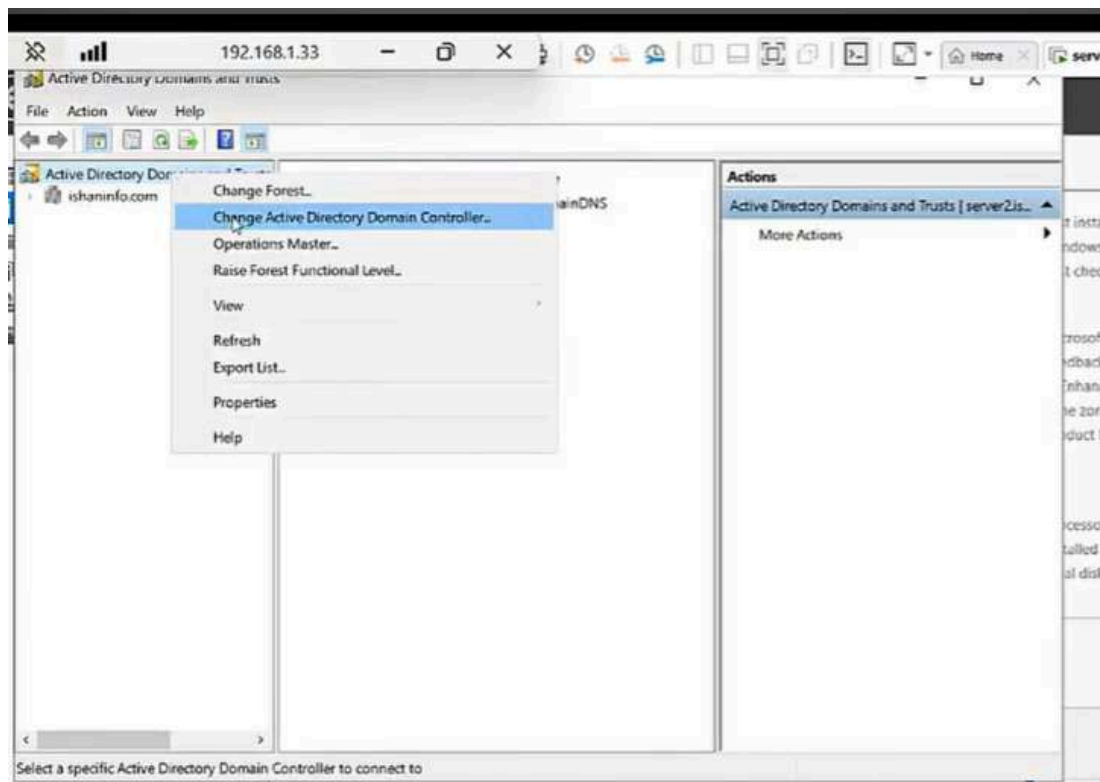
Transfer to server2

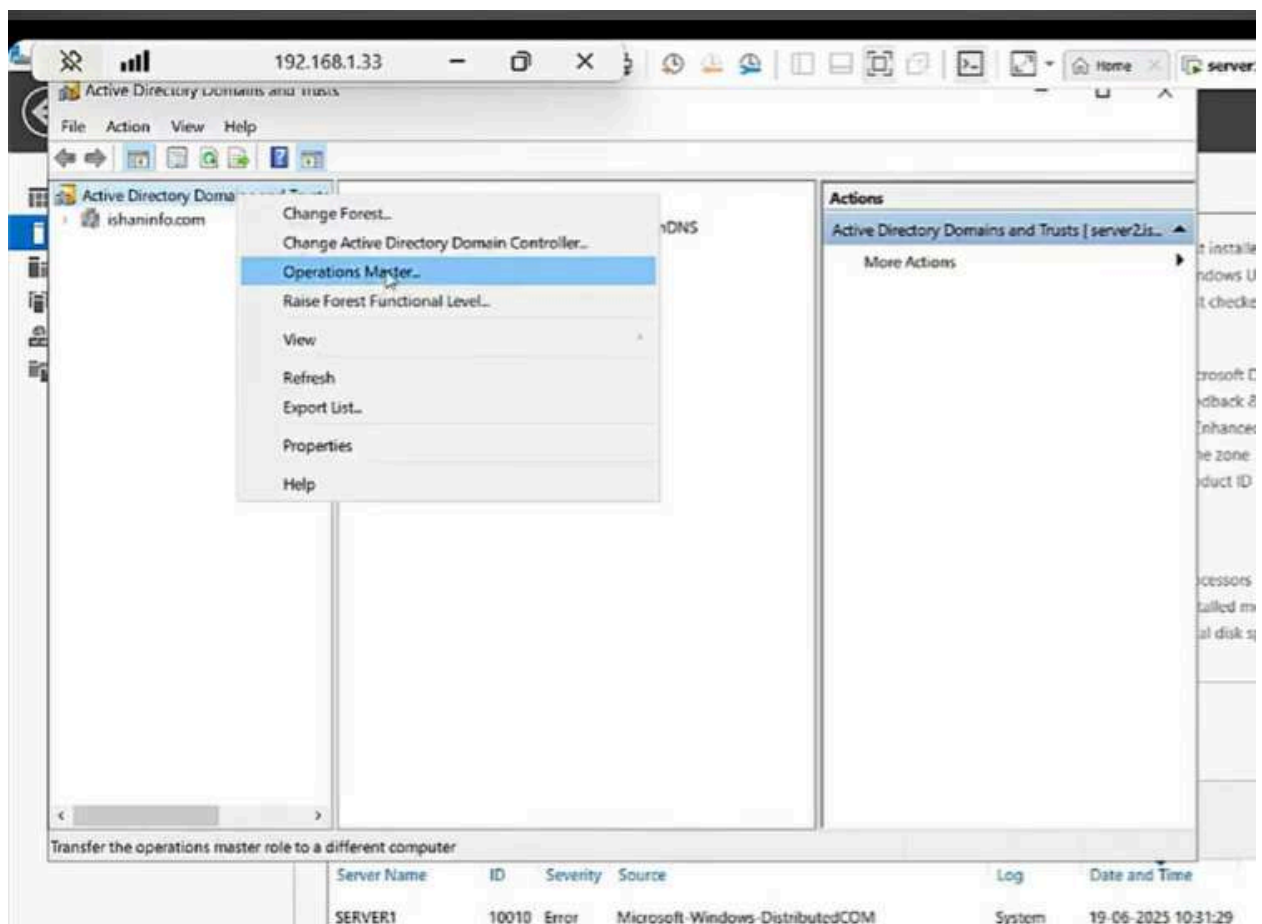
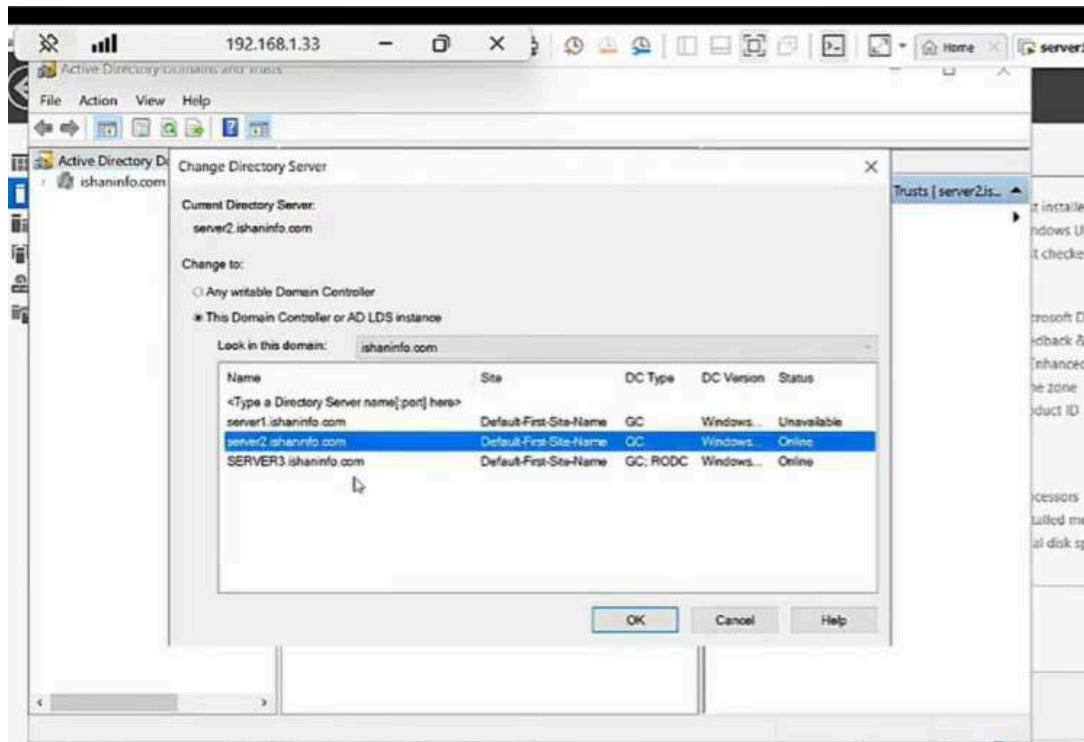


Three roles already set in server2

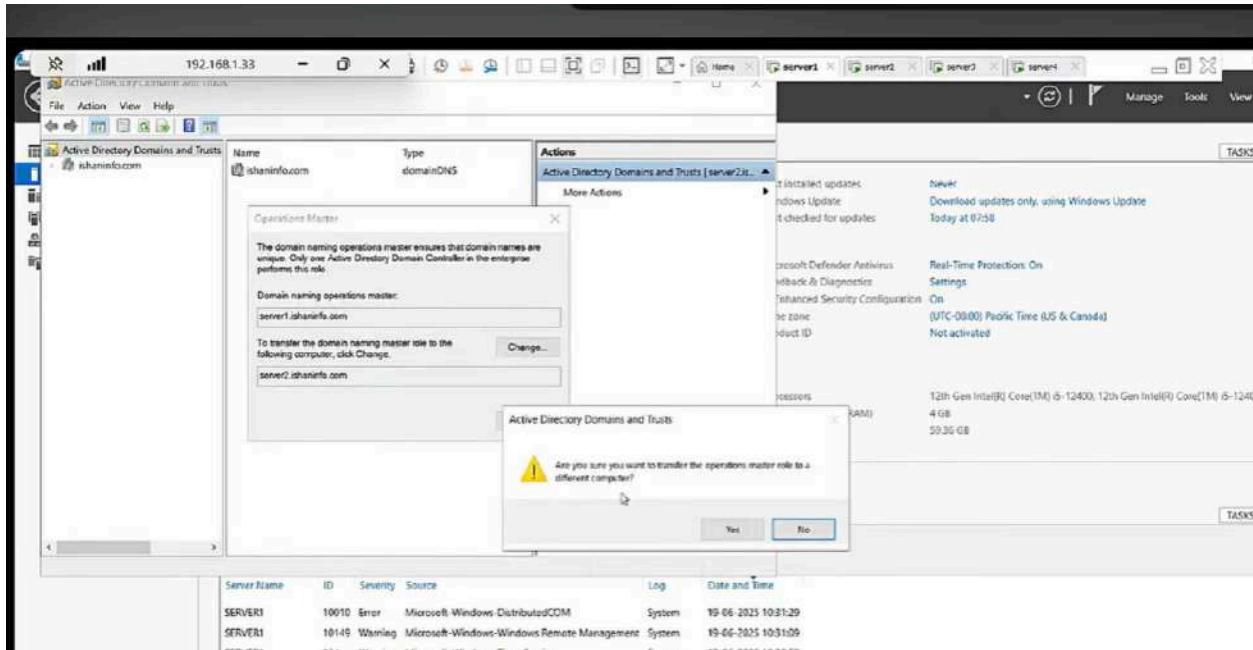


For transfer the roles to server2 Domain Naming Master we have to go Active Directory Domain and Trusts



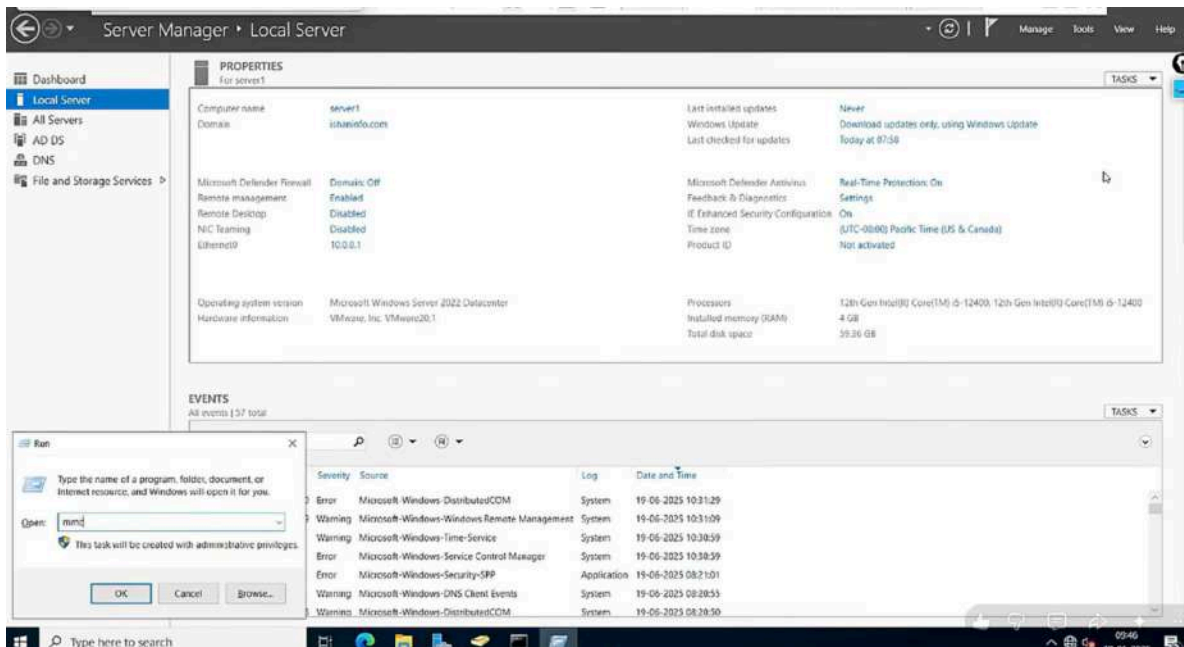


And change to server2

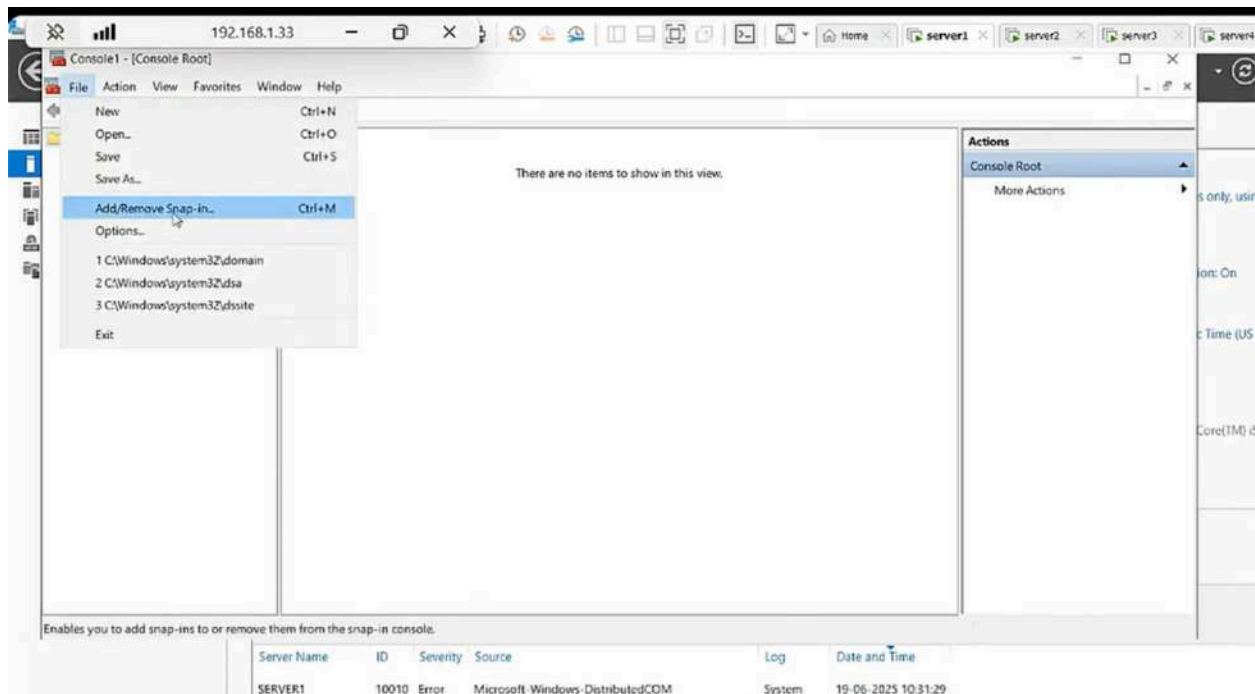


Now one role left which Schema Master

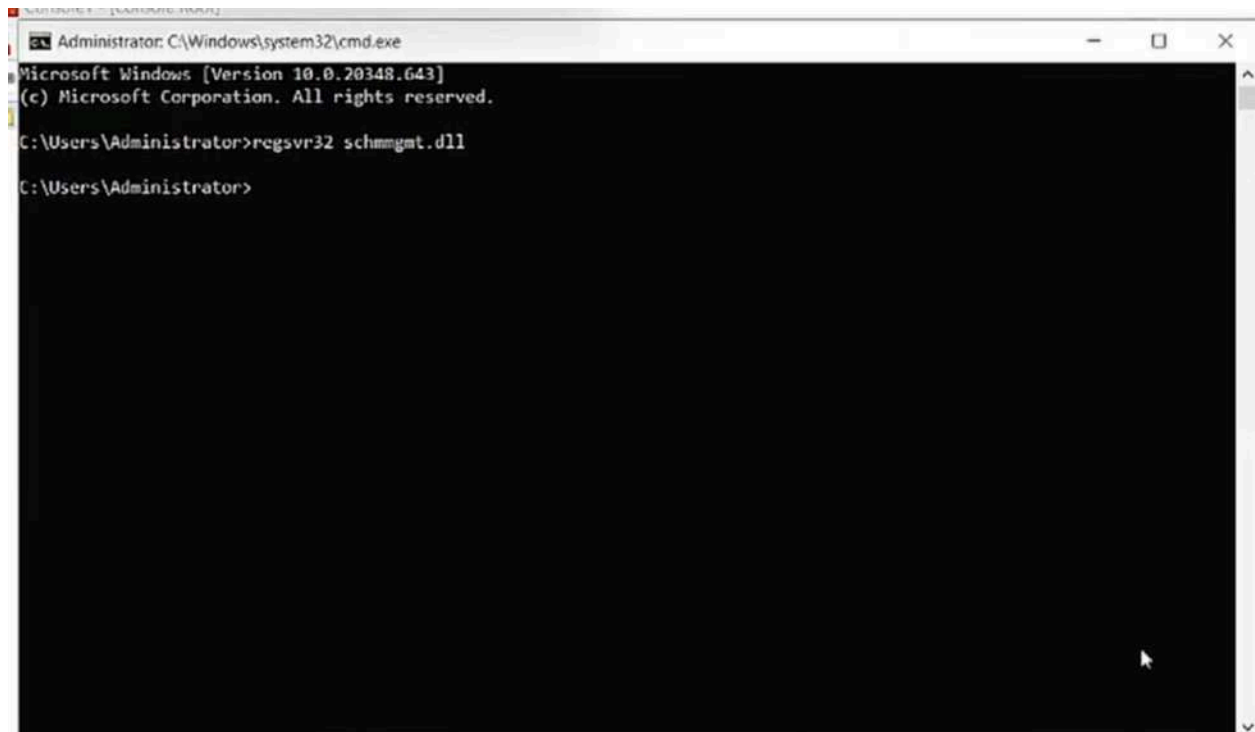
Now go to mmc

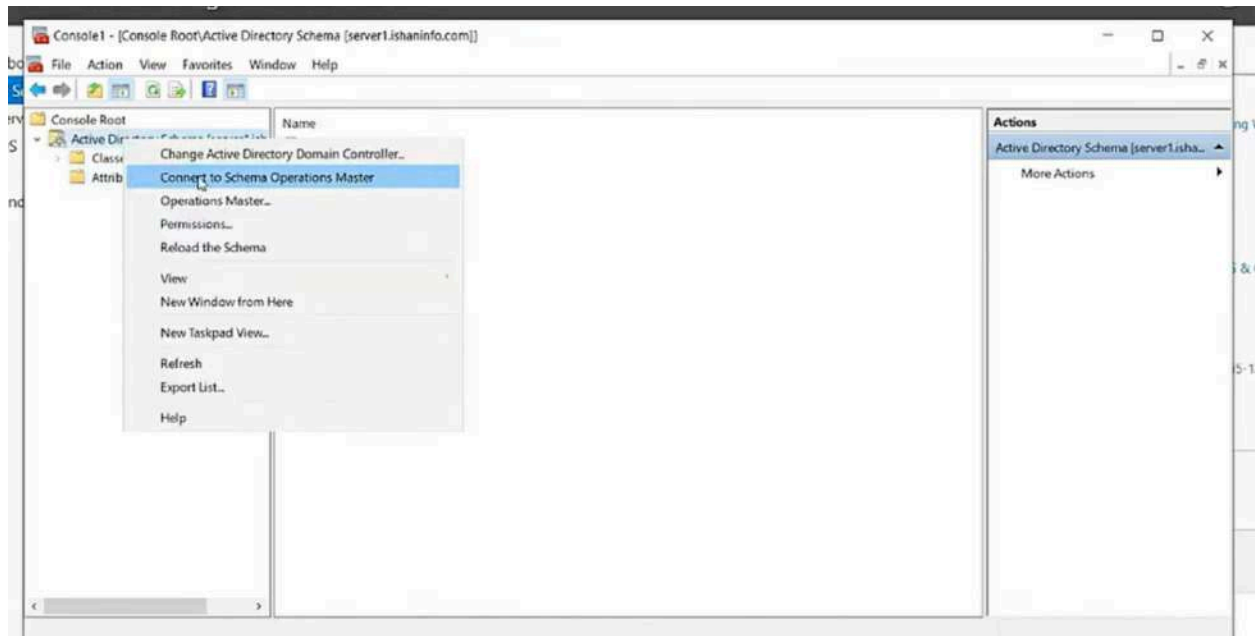


v



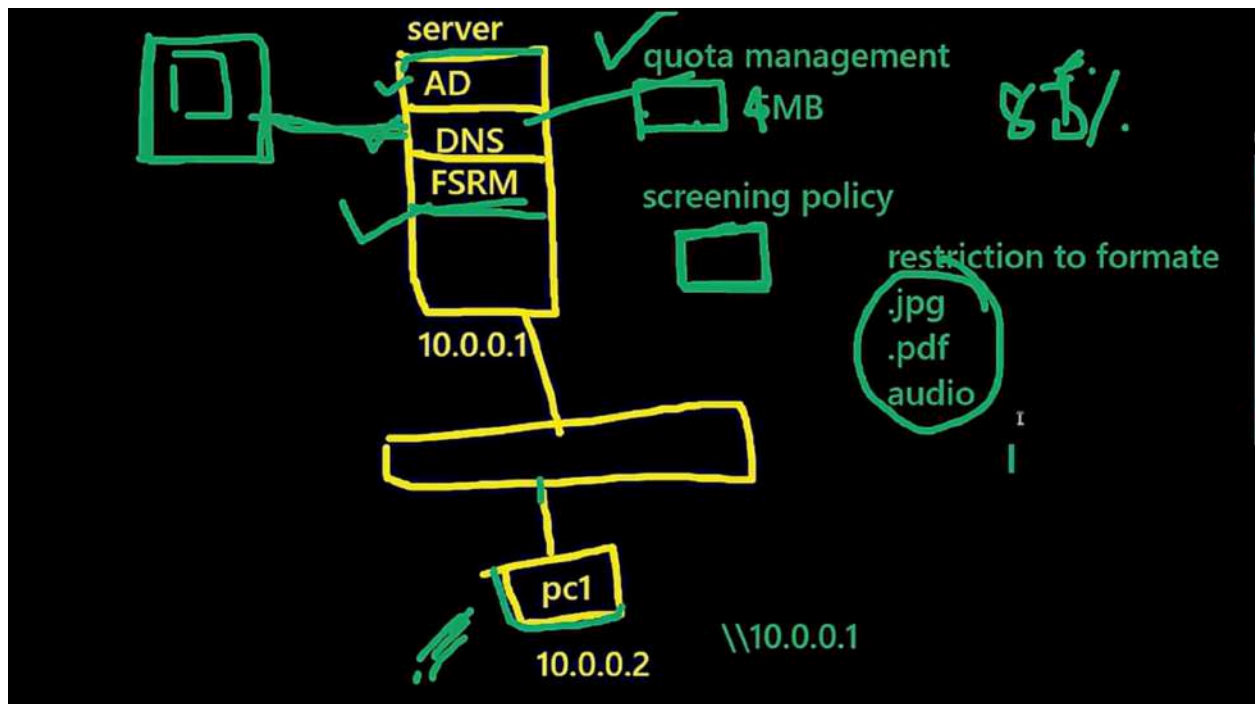
To import Schema so that we use this command

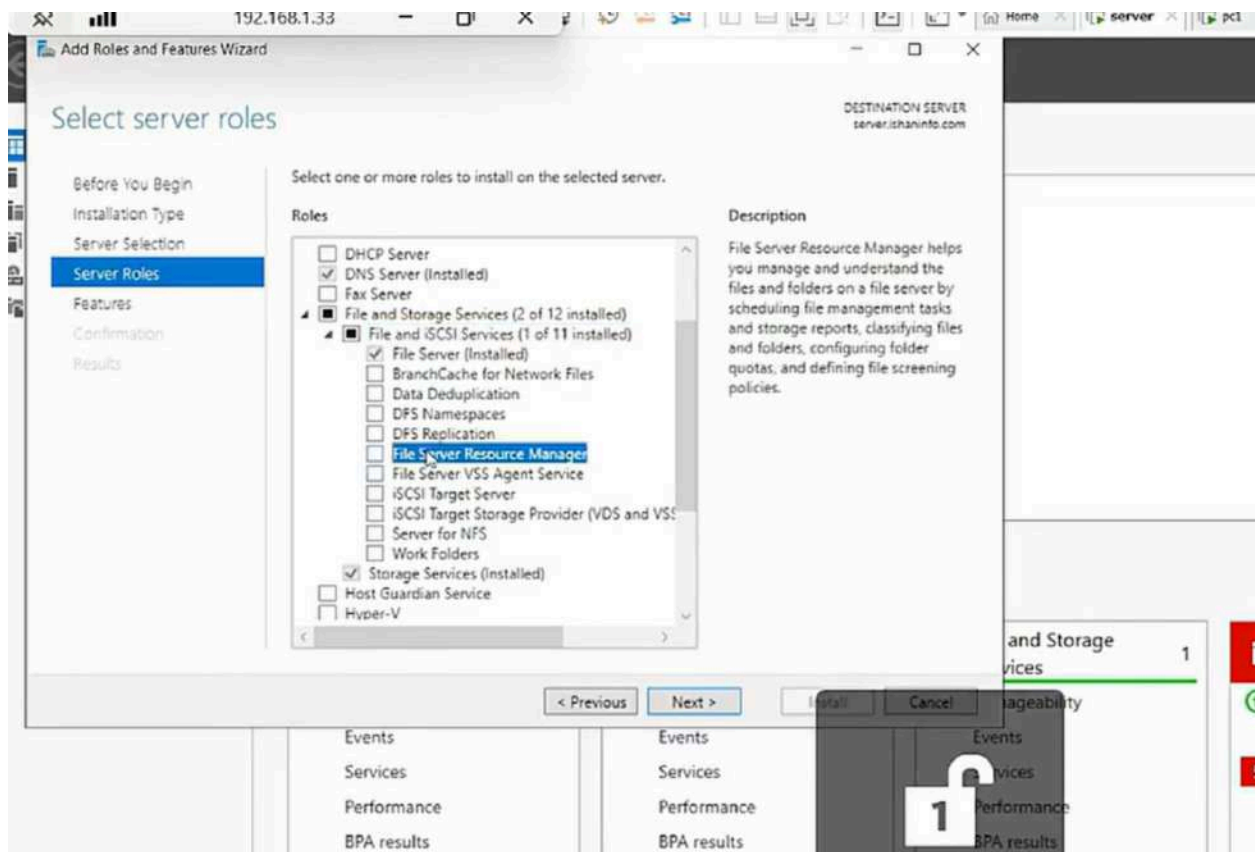
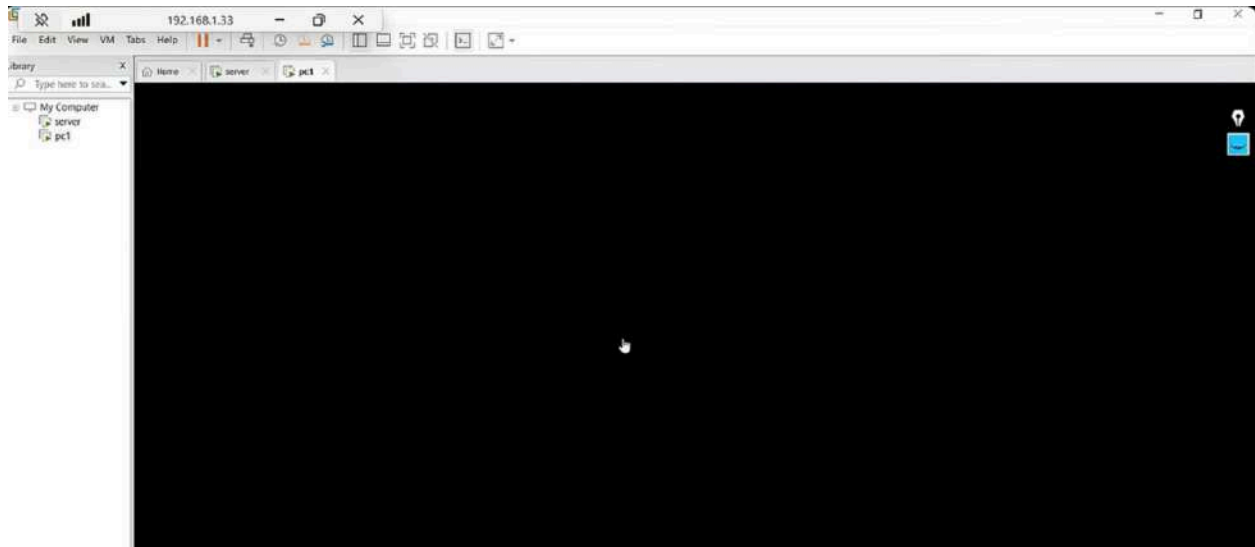


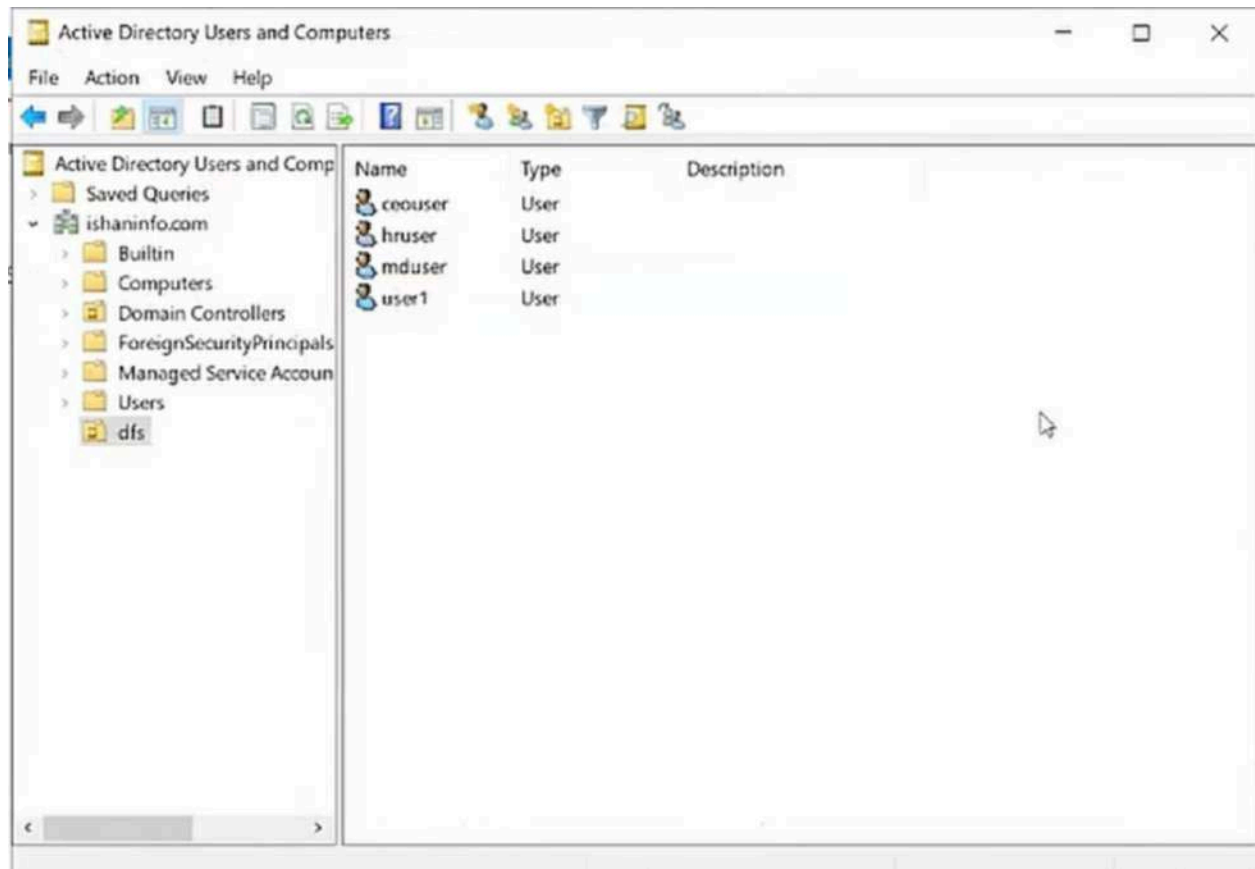
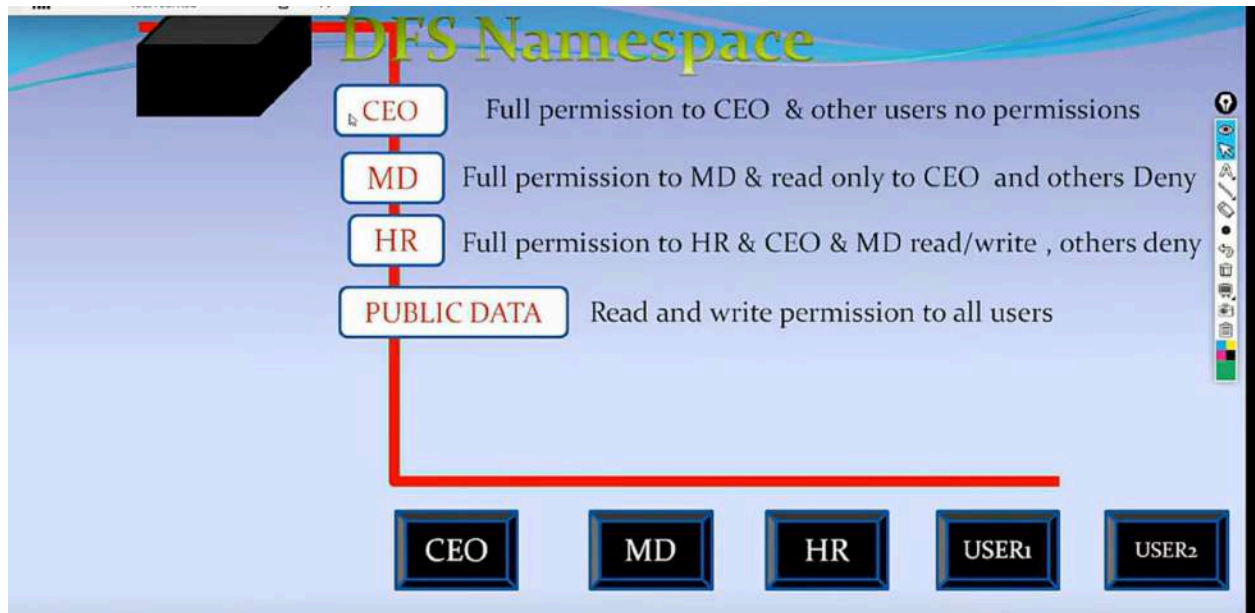


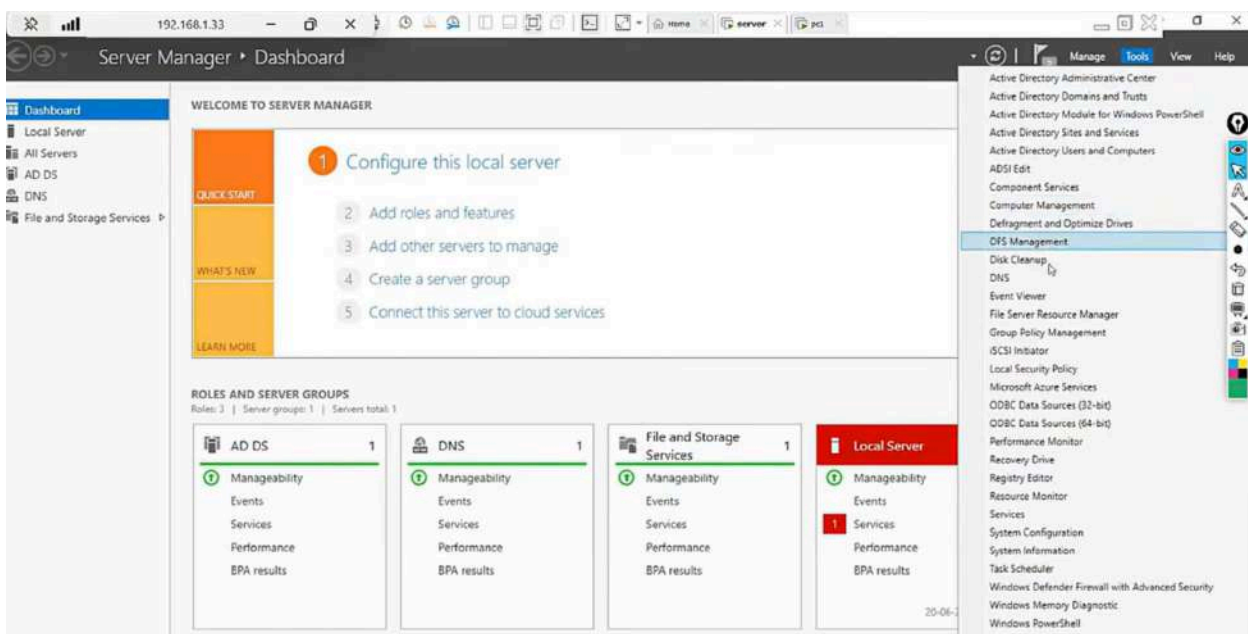
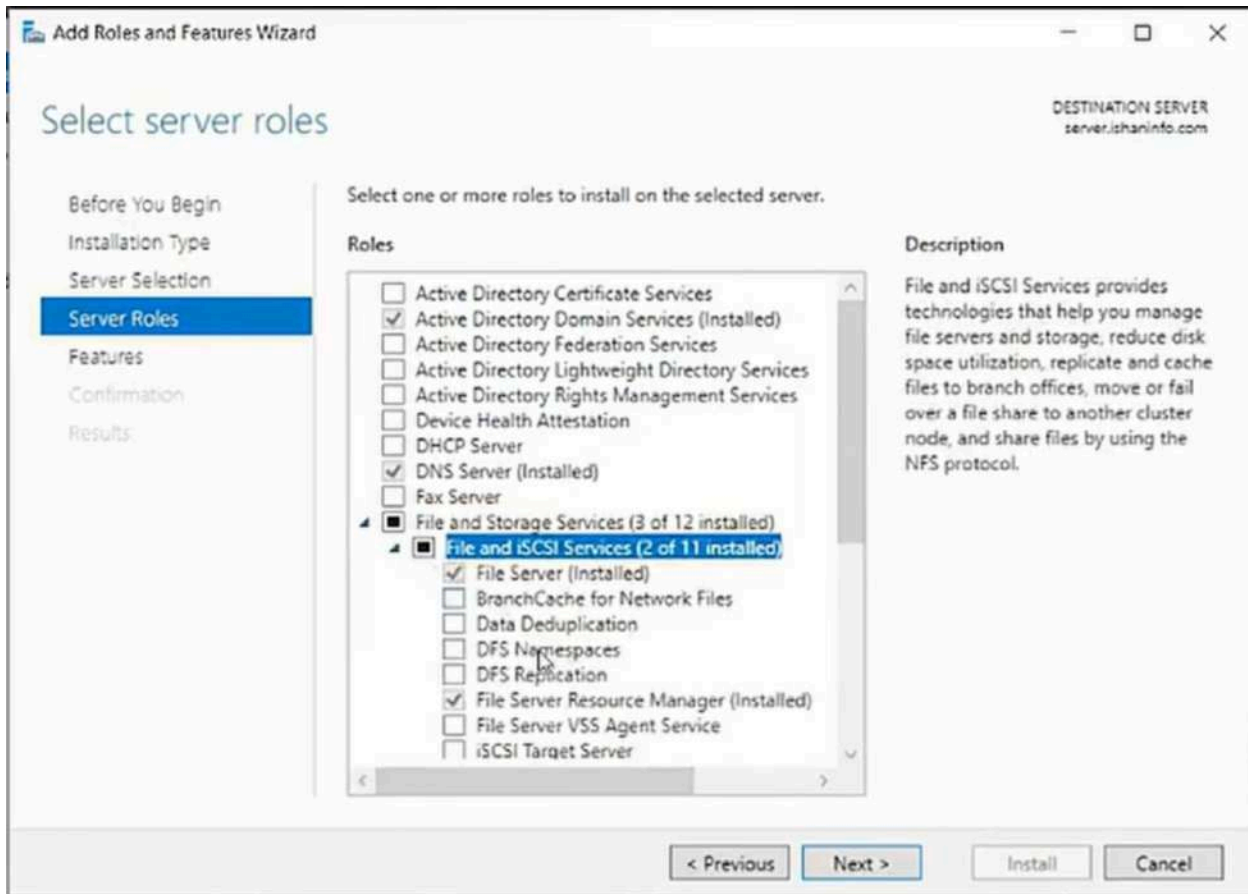
After all roles transfer to server2 then server becomes the Root Domain Controller or Primary Domain Controller

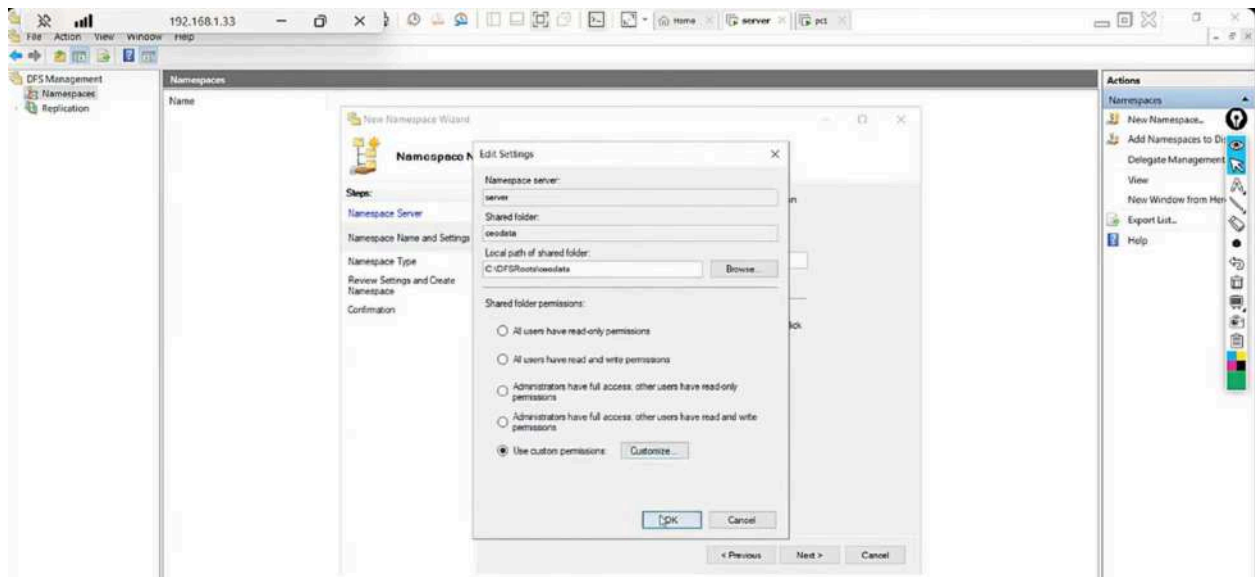
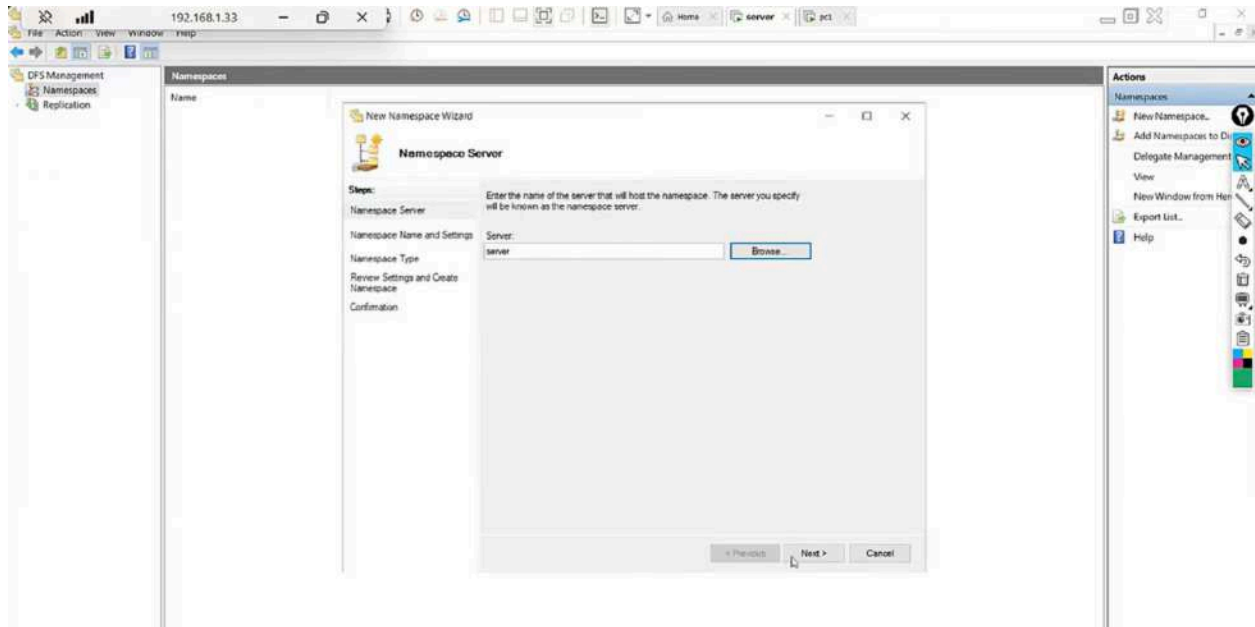
FSRM Setup:

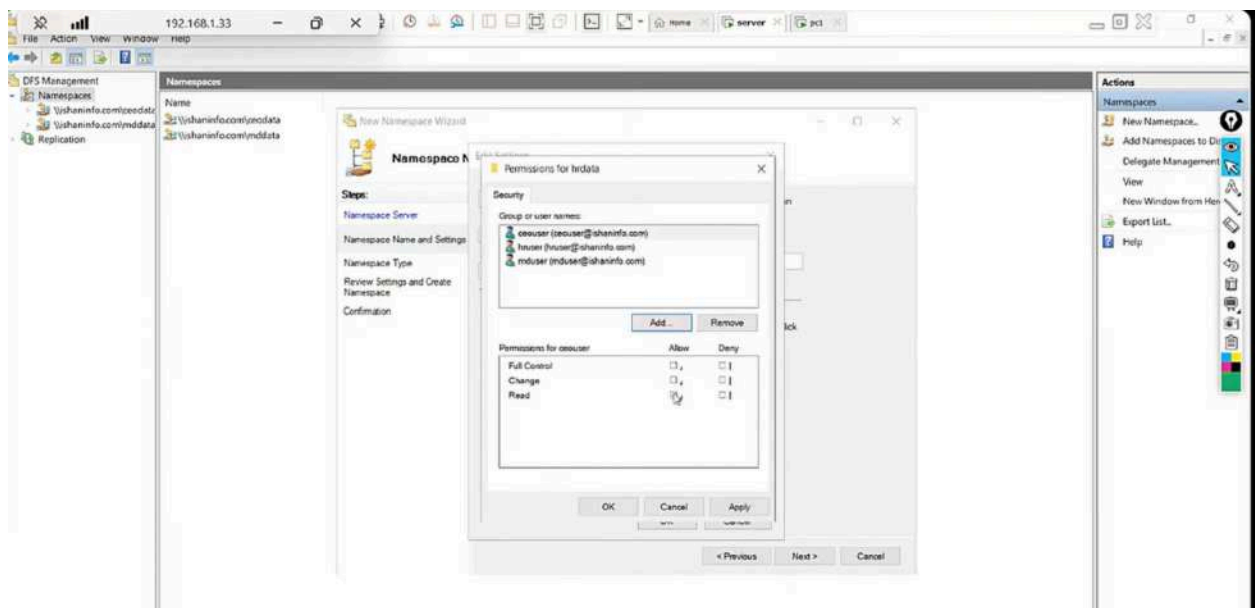
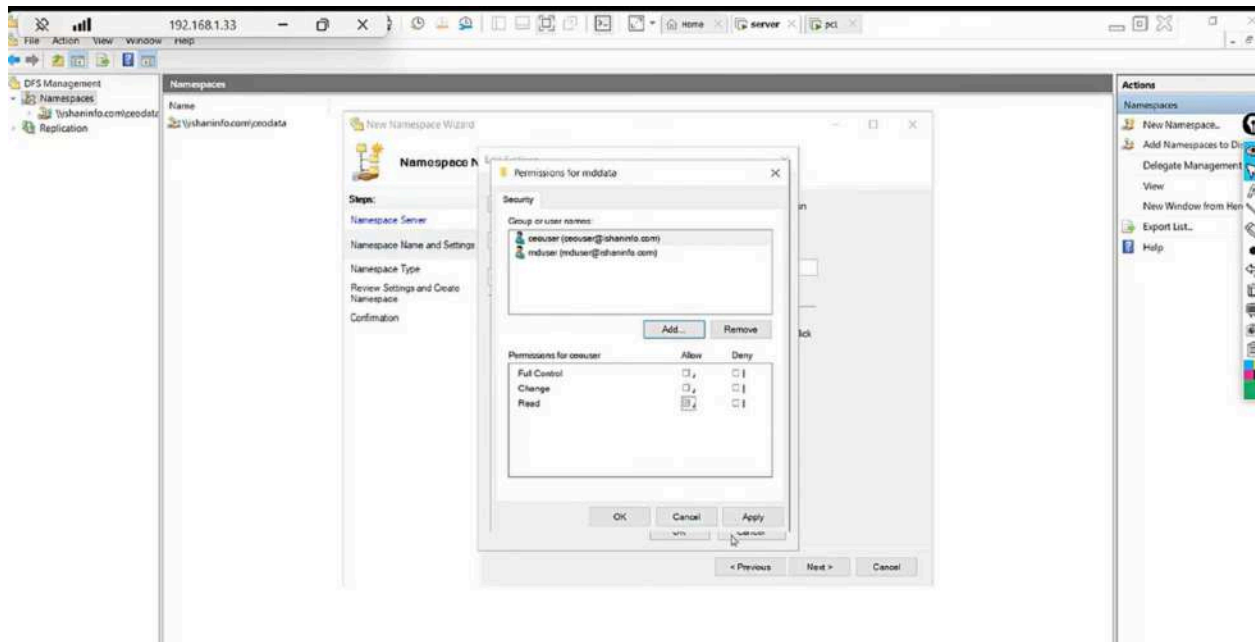




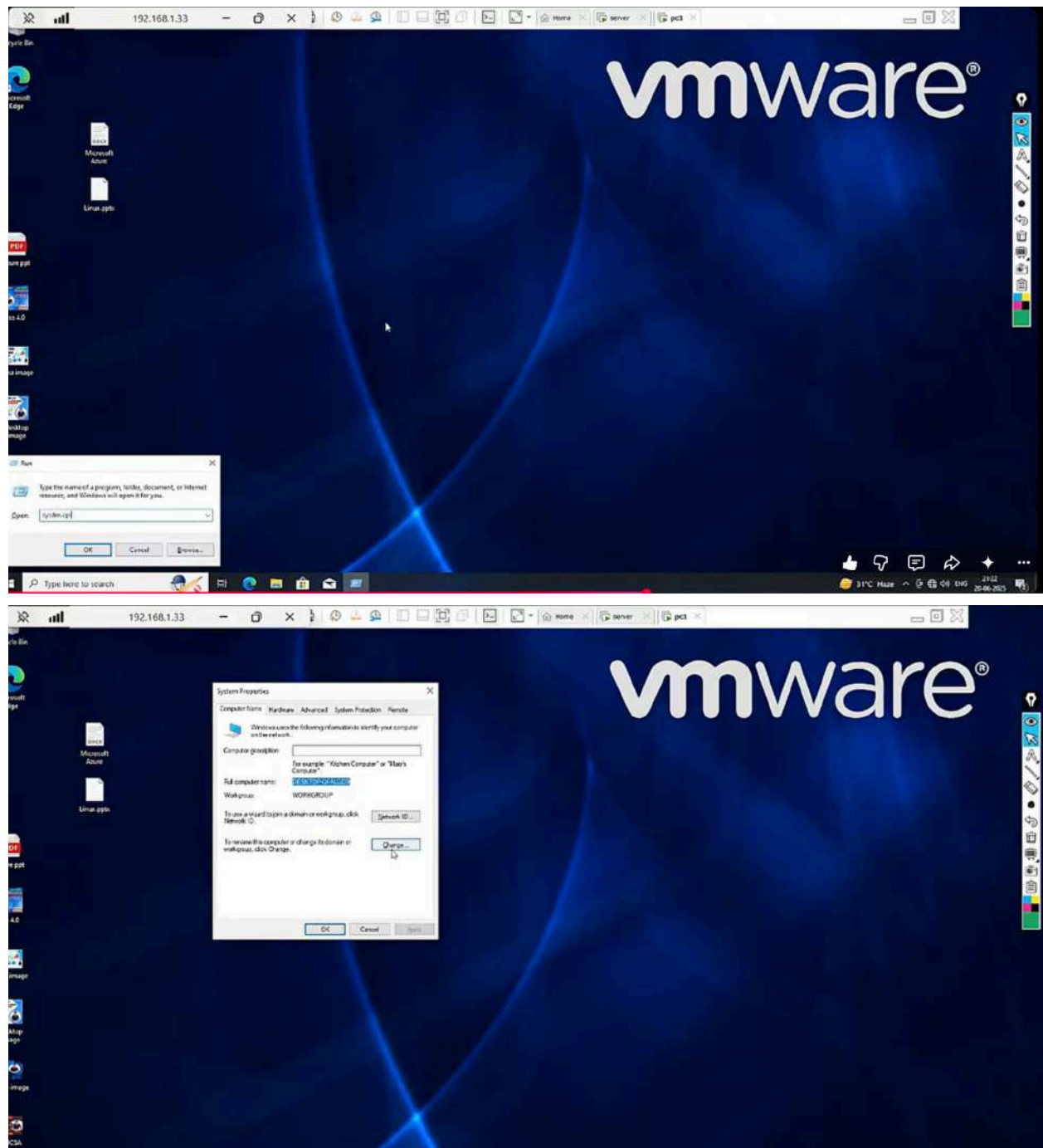


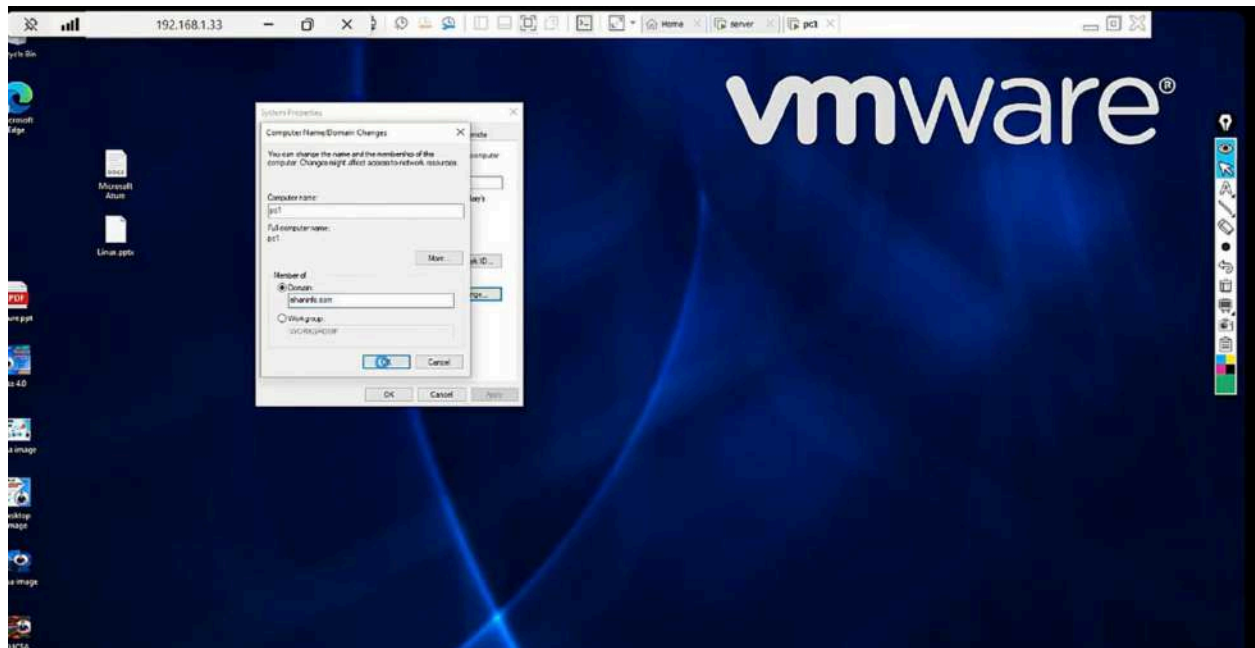






How to know this pc1 part of this server.
To check we have to type sysdm.cpl

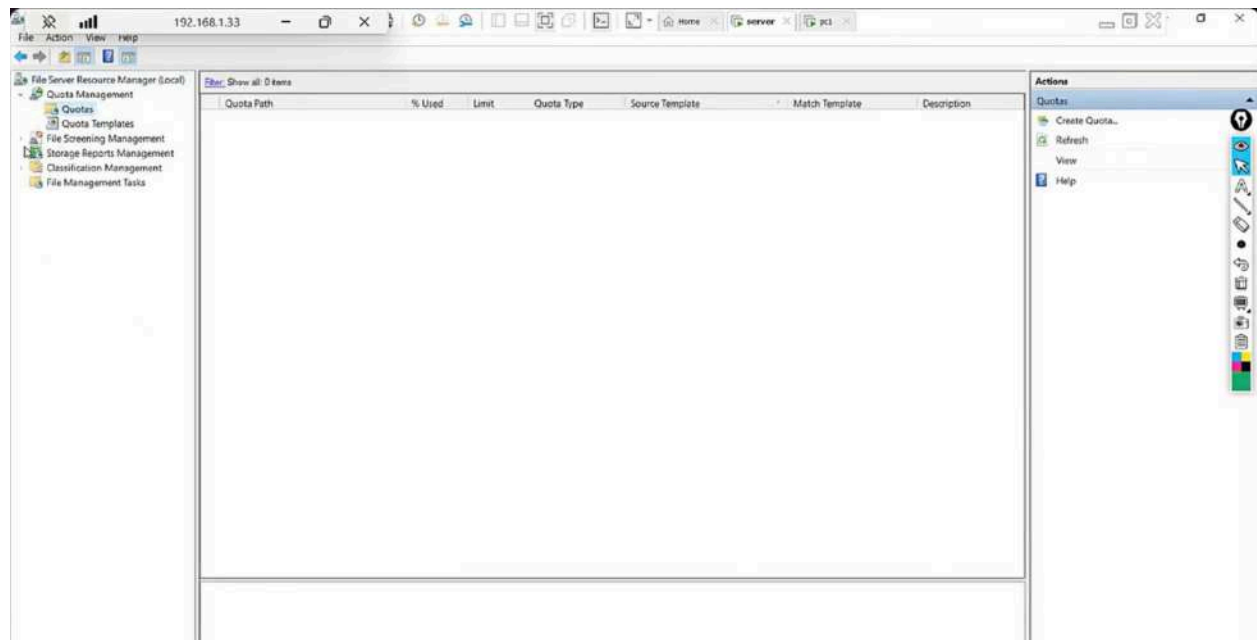
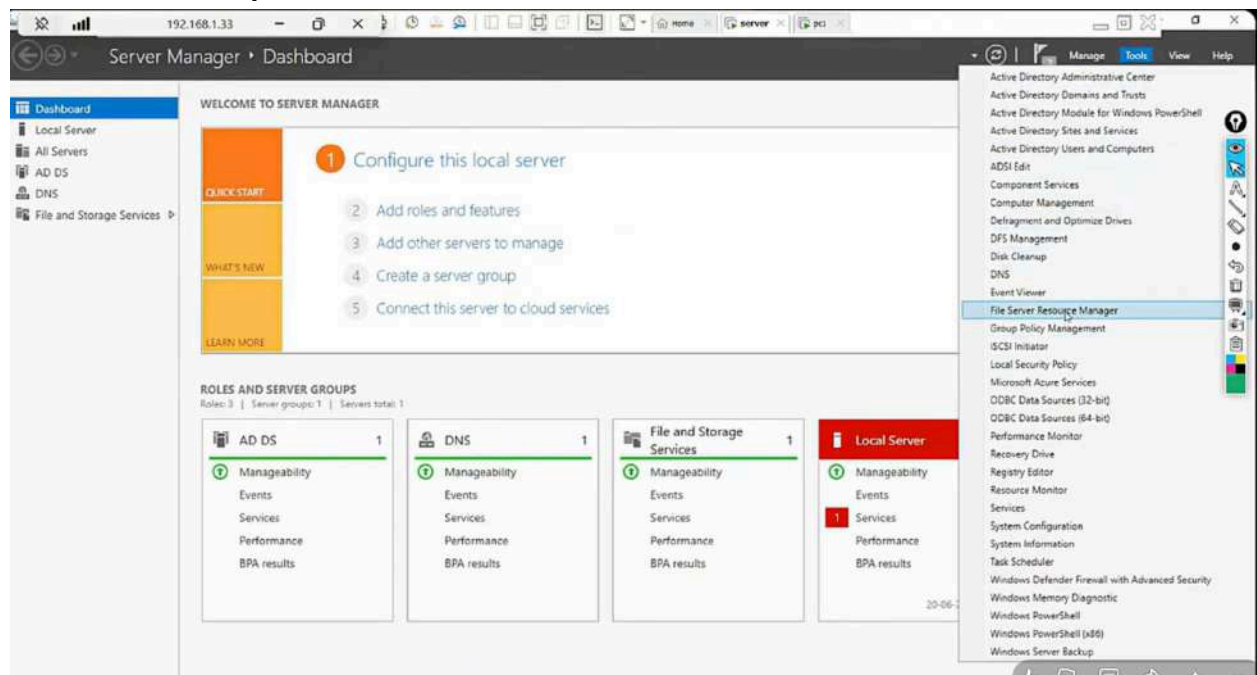




After that we can access server



Even we can set quota or even restrict



Windows server backup

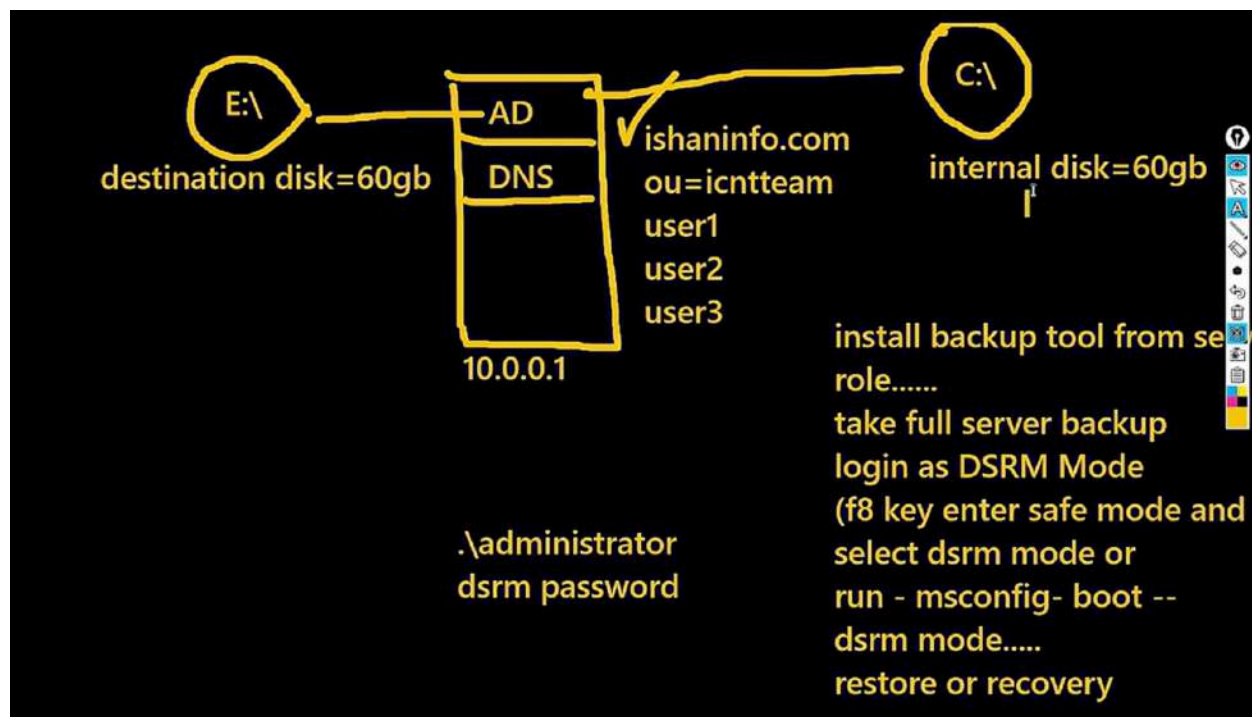
System recovery

Types of windows server backup

- File & folder backup
- Volume shadow copy backup
- System state backup
- Bare metal backup
- Full server backup

Windows server restore recovery

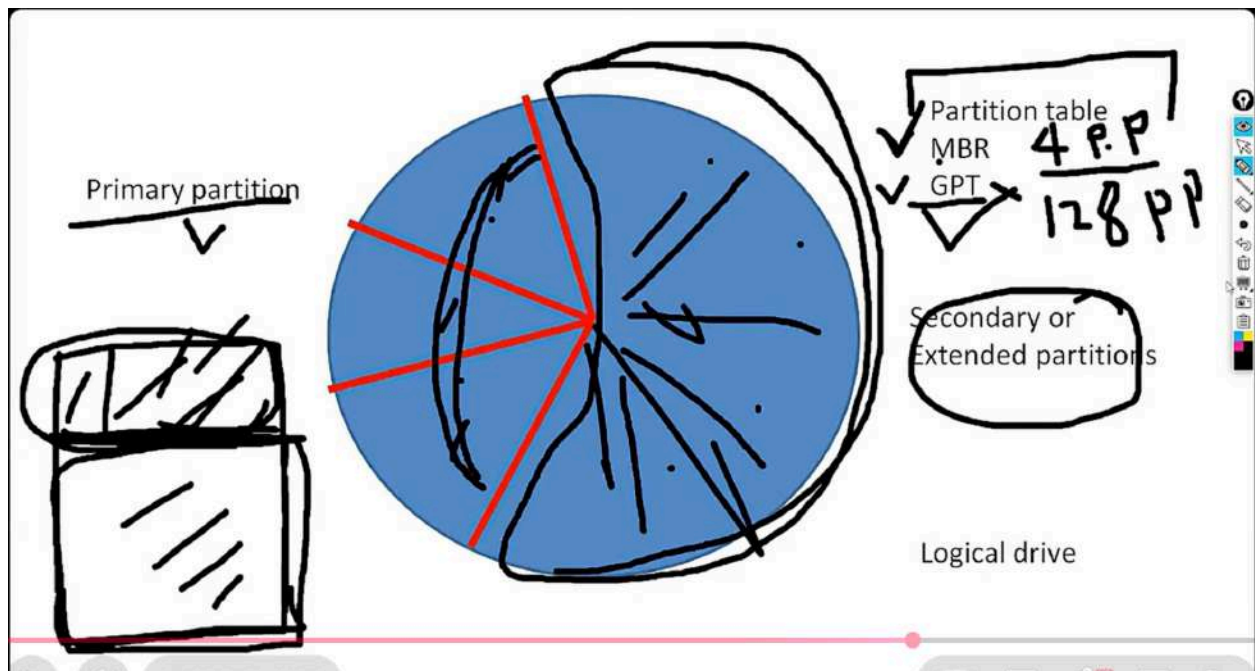
- File & folder (data)
- System state (DSRM)
- Disaster recovery



Storage

Disk Management

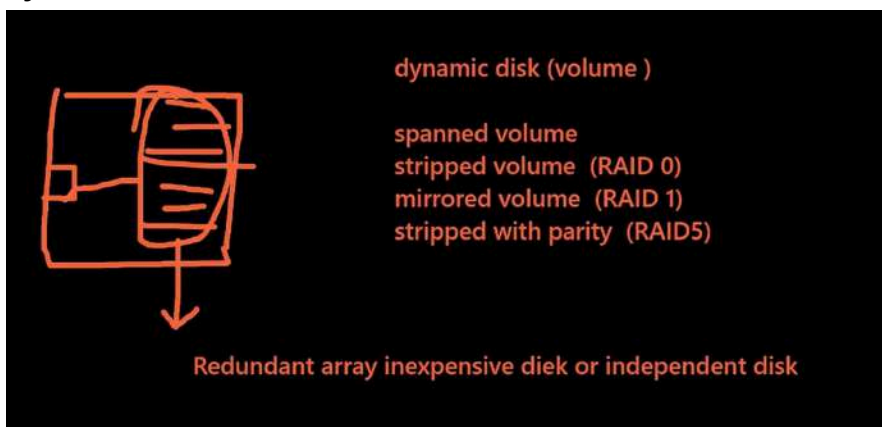
- Basic Disk
- Dynamic Disk



Bad sector check:

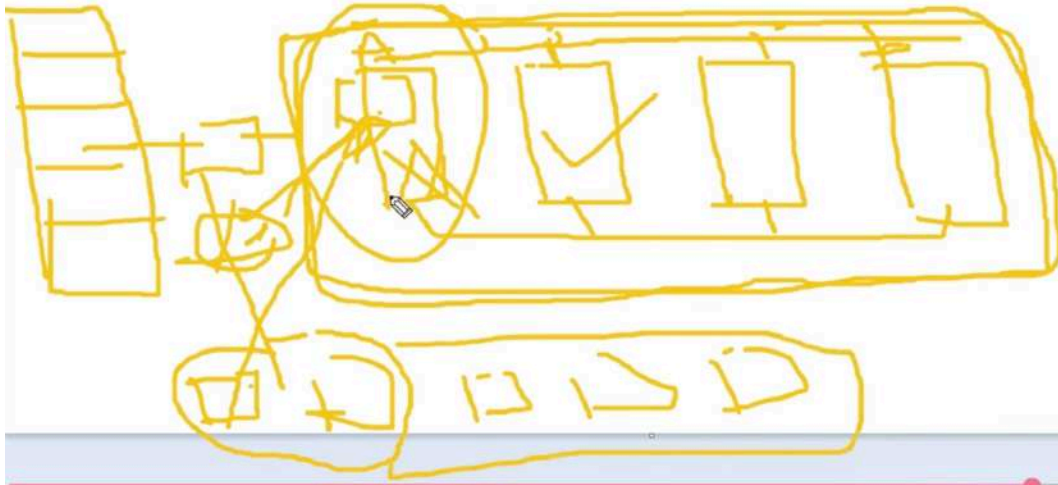


Dynamic Disk:

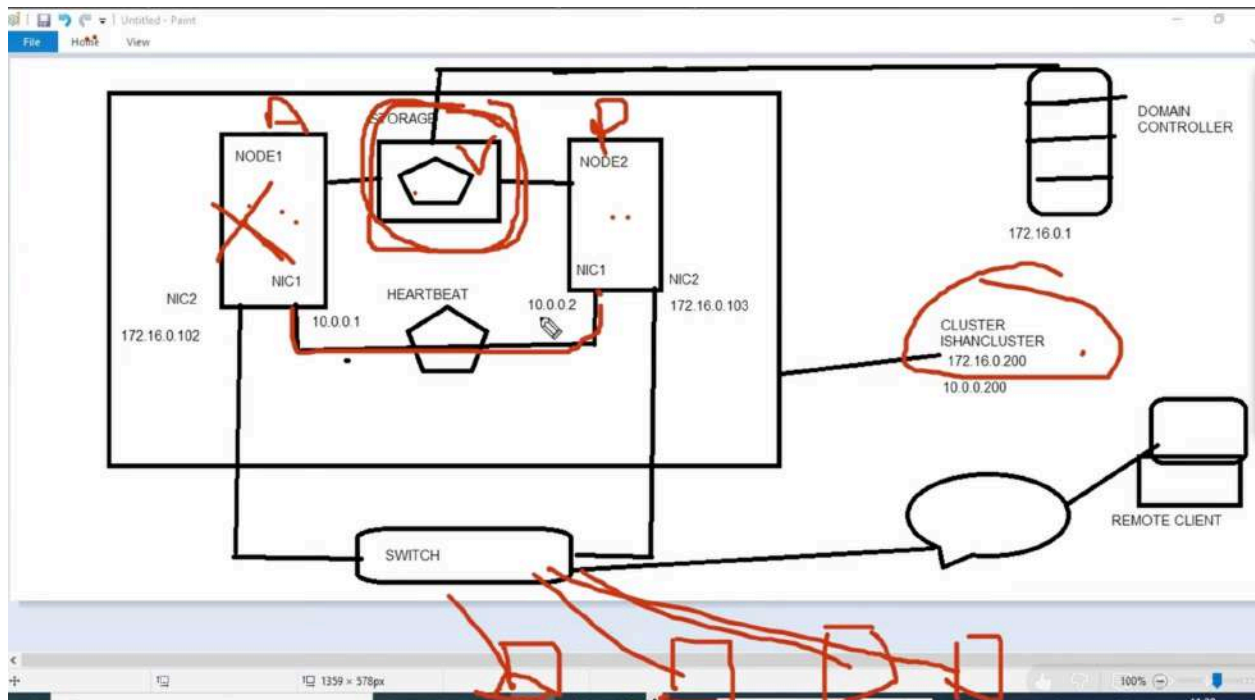


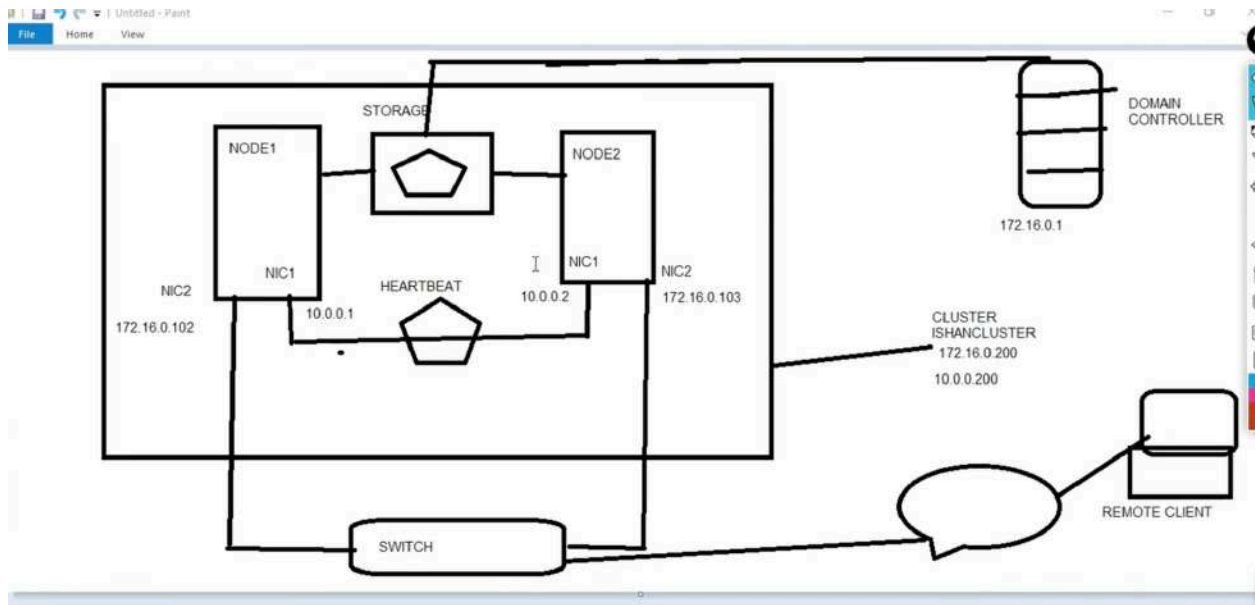
Hot swappable

Failover Clustering Lab



quorum disk





1. a failover cluster is a group of independent computers that work together to increase the availability and scalability of clustered roles (formerly called clustered applications and services).
2. the clustered servers (called nodes) are connected by physical cables and by software.
3. if one or more of the cluster nodes fail, other nodes begin to provide service (a process known as failover)
4. in addition, the clustered roles are proactively monitored to verify that they are working properly. if they are not working, they are restarted or moved to another node.
5. with the failover clustering feature users experience a minimum of disruptions in service.

